

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277885

Kind Code

A1

Publication Date

September 04, 2025

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REFLECTION-BASED MULTIPATH POSITION ESTIMATION

Abstract

Disclosed are techniques for wireless communication. In an aspect, a position estimation entity (e.g., UE, gNB, server, etc.) determines locations of multiple reflectors (e.g., buildings, etc.) based upon sensing operations by a wireless node (e.g., gNB, UE, etc.). The position estimation entity determines PRS configuration associated with multiple paths from the wireless node to a UE, some of the paths including reflections off of the multiple reflectors. The wireless node transmits and/or measures PRS(s) to and/or from the UE in accordance with the PRS configuration. The position estimation entity receives measurement information associated with the communication of the one or more PRSs. The position estimation entity derives a position estimate of the UE.

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Family ID: 84364177

Appl. No.: 18/702563

Filed (or PCT Filed): October 21, 2022

PCT No.: PCT/US2022/078502

Foreign Application Priority Data

GR 20210100801

Nov. 16, 2021

Publication Classification

Int. Cl.: G01S5/02 (20100101); G01S5/06 (20060101); H04W64/00 (20090101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present Application for Patent claims the benefit of GR application Ser. No. 20210100801, entitled “REFLECTION-BASED MULTIPATH POSITION ESTIMATION”, filed Nov. 16, 2021, and is a national stage application, filed under 35 U.S.C. § 371, of International Patent Application No. PCT/US2022/078502, entitled, “REFLECTION-BASED MULTIPATH POSITION ESTIMATION”, filed Oct. 21, 2022, both of which are assigned to the assignee hereof and are expressly incorporated herein by reference in their entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] Aspects of the disclosure relate generally to wireless communications.

2. Description of the Related Art

[0003] Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service and a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax). There are presently many different types of wireless communication systems in use, including cellular and personal communications service (PCS) systems. Examples of known cellular systems include the cellular analog advanced mobile phone system (AMPS), and digital cellular systems based on code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), the Global System for Mobile communications (GSM), etc.

[0004] A fifth generation (5G) wireless standard, referred to as New Radio (NR), enables higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide higher data rates as compared to previous standards, more accurate positioning (e.g., based on reference signals for positioning (RS-P), such as downlink, uplink, or sidelink positioning reference signals (PRS)), and other technical enhancements. These enhancements, as well as the use of higher frequency bands, advances in PRS processes and technology, and high-density deployments for 5G, enable highly accurate 5G-based positioning.

SUMMARY

[0005] The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

[0006] In an aspect, a method of operating a position estimation entity includes determining a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determining a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determining a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with

reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmitting an indication of the PRS configuration to the wireless node, the UE, or a combination thereof, receiving third measurement information associated with the position estimation session; and deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0007] In an aspect, a method of operating a wireless node includes performing a first sensing operation associated with a first reflector; performing a second sensing operation associated with a second reflector; reporting, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receiving, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmitting or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtaining third measurement information associated with the position estimation session.

[0008] In an aspect, a position estimation entity includes a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: determine a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determine a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determine a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit, via the at least one transceiver, an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receive, via the at least one transceiver, third measurement information associated with the position estimation session; and derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0009] In an aspect, a wireless node includes a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: perform a first sensing operation associated with a first reflector; perform a second sensing operation associated with a second reflector; report, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receive, via the at least one transceiver, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit, via the at least one transceiver, or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtain

third measurement information associated with the position estimation session.

[0010] In an aspect, a position estimation entity includes means for determining a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; means for determining a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; means for determining a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; means for transmitting an indication of the PRS configuration to the wireless node, the UE, or a combination thereof, means for receiving third measurement information associated with the position estimation session; and means for deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0011] In an aspect, a wireless node includes means for performing a first sensing operation associated with a first reflector; means for performing a second sensing operation associated with a second reflector; means for reporting, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; means for receiving, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; means for transmitting or measuring one or more PRSs with the UE in accordance with the PRS configuration; and means for obtaining third measurement information associated with the position estimation session.

[0012] In an aspect, a non-transitory computer-readable medium storing computer-executable instructions that, when executed by a position estimation entity, cause the position estimation entity to: determine a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determine a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determine a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receive third measurement information associated with the position estimation session; and derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0013] In an aspect, a non-transitory computer-readable medium storing computer-executable instructions that, when executed by a wireless node, cause the wireless node to: perform a first sensing operation associated with a first reflector; perform a second sensing operation associated with a second reflector; report, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receive, in response to the report of the first measurement information

and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtain third measurement information associated with the position estimation session.

[0014] Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects and not limitation thereof.

[0016] FIG. 1 illustrates an example wireless communications system, according to aspects of the disclosure.

[0017] FIGS. 2A and 2B illustrate example wireless network structures, according to aspects of the disclosure.

[0018] FIGS. 3A, 3B, and 3C are simplified block diagrams of several sample aspects of components that may be employed in a user equipment (UE), a base station, and a network entity, respectively, and configured to support communications as taught herein.

[0019] FIG. 4 is a diagram illustrating an example frame structure, according to aspects of the disclosure.

[0020] FIG. 5 is a diagram illustrating various downlink channels within an example downlink slot, according to aspects of the disclosure.

[0021] FIG. 6 is a diagram illustrating various uplink channels within an example uplink slot, according to aspects of the disclosure.

[0022] FIG. 7 illustrates time and frequency resources used for sidelink communication.

[0023] FIG. 8 is a diagram of an example positioning reference signal (PRS) configuration for the PRS transmissions of a given base station, according to aspects of the disclosure.

[0024] FIG. 9 is a diagram illustrating an example downlink positioning reference signal (DL-PRS) configuration for two transmission-reception points (TRPs) operating in the same positioning frequency layer, according to aspects of the disclosure.

[0025] FIG. 10 is a graph representing a radio frequency (RF) channel impulse response over time, according to aspects of the disclosure.

[0026] FIG. 11 illustrates examples of various positioning methods supported in New Radio (NR), according to aspects of the disclosure.

[0027] FIG. 12 is a diagram illustrating an example round-trip-time (RTT) procedure for determining a location of a UE, according to aspects of the disclosure.

[0028] FIG. 13 is a diagram showing example timings of RTT measurement signals exchanged between a base station and a UE, according to aspects of the disclosure.

[0029] FIG. 14 is a diagram illustrating example timings of RTT measurement signals exchanged between a base station and a UE, according to aspects of the disclosure.

[0030] FIG. 15 illustrates a time difference of arrival (TDOA)-based positioning procedure in an example wireless communications system, according to aspects of the disclosure.

[0031] FIG. 16 is a diagram illustrating an example base station in communication with an example UE, according to aspects of the disclosure.

[0032] FIG. **17** illustrates an exemplary process of communication according to aspects of the disclosure.

[0033] FIG. **18** illustrates an exemplary process of communication according to aspects of the disclosure.

[0034] FIG. **19** illustrates a position estimation environment in accordance with an example implementation of the processes of FIGS. **17-18**, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0035] Aspects of the disclosure are provided in the following description and related drawings directed to various examples provided for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.

[0036] The words “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.

[0037] Those of skill in the art will appreciate that the information and signals described below may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description below may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.

[0038] Further, many aspects are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits (ASICs)), by program instructions being executed by one or more processors, or by a combination of both. Additionally, the sequence(s) of actions described herein can be considered to be embodied entirely within any form of non-transitory computer-readable storage medium having stored therein a corresponding set of computer instructions that, upon execution, would cause or instruct an associated processor of a device to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, all of which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the aspects described herein, the corresponding form of any such aspects may be described herein as, for example, “logic configured to” perform the described action.

[0039] As used herein, the terms “user equipment” (UE) and “base station” are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset locating device, wearable (e.g., smartwatch, glasses, augmented reality (AR)/virtual reality (VR) headset, etc.), vehicle (e.g., automobile, motorcycle, bicycle, etc.), Internet of Things (IoT) device, etc.) used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or “UT,” a “mobile device,” a “mobile terminal,” a “mobile station,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible

for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 specification, etc.) and so on.

[0040] A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB, an evolved NodeB (CNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

[0041] The term “base station” may refer to a single physical transmission-reception point (TRP) or to multiple physical TRPs that may or may not be co-located. For example, where the term “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

[0042] In some implementations that support positioning of UEs, a base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

[0043] An “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

[0044] FIG. 1 illustrates an example wireless communications system **100**, according to aspects of the disclosure. The wireless communications system **100** (which may also be referred to as a wireless wide area network (WWAN)) may include various base stations **102** (labeled “BS”) and various UEs **104**. The base stations **102** may include macro cell base stations (high power cellular

base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base stations may include eNBs and/or ng-eNBs where the wireless communications system **100** corresponds to an LTE network, or gNBs where the wireless communications system **100** corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

[0045] The base stations **102** may collectively form a RAN and interface with a core network **170** (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links **122**, and through the core network **170** to one or more location servers **172** (e.g., a location management function (LMF) or a secure user plane location (SUPL) location platform (SLP)). The location server(s) **172** may be part of core network **170** or may be external to core network **170**. A location server **172** may be integrated with a base station **102**. A UE **104** may communicate with a location server **172** directly or indirectly. For example, a UE **104** may communicate with a location server **172** via the base station **102** that is currently serving that UE **104**. A UE **104** may also communicate with a location server **172** through another path, such as via an application server (not shown), via another network, such as via a wireless local area network (WLAN) access point (AP) (e.g., AP **150** described below), and so on. For signaling purposes, communication between a UE **104** and a location server **172** may be represented as an indirect connection (e.g., through the core network **170**, etc.) or a direct connection (e.g., as shown via direct connection **128**), with the intervening nodes (if any) omitted from a signaling diagram for clarity.

[0046] In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate with each other directly or indirectly (e.g., through the EPC/5GC) over backhaul links **134**, which may be wired or wireless.

[0047] The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each geographic coverage area **110**. A “cell” is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), an enhanced cell identifier (ECI), a virtual cell identifier (VCI), a cell global identifier (CGI), etc.) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term “cell” may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms “cell” and “TRP” may be used interchangeably. In some cases, the term “cell” may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.

[0048] While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102'** (labeled “SC” for “small cell”) may have a geographic coverage area **110'** that substantially overlaps with the geographic coverage area **110** of one or more macro cell base

stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

[0049] The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

[0050] The wireless communications system **100** may further include a wireless local area network (WLAN) access point (AP) **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 GHz). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available.

[0051] The small cell base station **102'** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102'** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102'**, employing LTE/5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

[0052] The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and the UE **182** may utilize beamforming (transmit and/or receive) over a mmW communication link **184** to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations **102** may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

[0053] Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while cancelling to

suppress radiation in undesired directions.

[0054] Transmit beams may be quasi-co-located, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically co-located. In NR, there are four types of quasi-co-location (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

[0055] In receive beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain in that direction of all other receive beams available to the receiver. This results in a stronger received signal strength (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

[0056] Transmit and receive beams may be spatially related. A spatial relation means that parameters for a second beam (e.g., a transmit or receive beam) for a second reference signal can be derived from information about a first beam (e.g., a receive beam or a transmit beam) for a first reference signal. For example, a UE may use a particular receive beam to receive a reference downlink reference signal (e.g., synchronization signal block (SSB)) from a base station. The UE can then form a transmit beam for sending an uplink reference signal (e.g., sounding reference signal (SRS)) to that base station based on the parameters of the receive beam.

[0057] Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

[0058] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

[0059] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies.

Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0060] With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band.

[0061] In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104/182** and the cell in which the UE **104/182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that different UEs **104/182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104/182** at any time. This is done, for example, to balance the load on different carriers. Because a “serving cell” (whether a PCell or an SCell) corresponds to a carrier frequency/component carrier over which some base station is communicating, the term “cell,” “serving cell,” “component carrier,” “carrier frequency,” and the like can be used interchangeably.

[0062] For example, still referring to FIG. **1**, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or “PCell”) and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers (“SCells”). The simultaneous transmission and/or reception of multiple carriers enables the UE **104/182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.

[0063] The wireless communications system **100** may further include a UE **164** that may communicate with a macro cell base station **102** over a communication link **120** and/or the mmW base station **180** over a mmW communication link **184**. For example, the macro cell base station **102** may support a PCell and one or more SCells for the UE **164** and the mm W base station **180** may support one or more SCells for the UE **164**.

[0064] In some cases, the UE **164** and the UE **182** may be capable of sidelink communication. Sidelink-capable UEs (SL-UEs) may communicate with base stations **102** over communication links **120** using the Uu interface (i.e., the air interface between a UE and a base station). SL-UEs (e.g., UE **164**, UE **182**) may also communicate directly with each other over a wireless sidelink **160**

using the PC5 interface (i.e., the air interface between sidelink-capable UEs). A wireless sidelink (or just “sidelink”) is an adaptation of the core cellular (e.g., LTE, NR) standard that allows direct communication between two or more UEs without the communication needing to go through a base station. Sidelink communication may be unicast or multicast, and may be used for device-to-device (D2D) media-sharing, vehicle-to-vehicle (V2V) communication, vehicle-to-everything (V2X) communication (e.g., cellular V2X (cV2X) communication, enhanced V2X (eV2X) communication, etc.), emergency rescue applications, etc. One or more of a group of SL-UEs utilizing sidelink communications may be within the geographic coverage area **110** of a base station **102**. Other SL-UEs in such a group may be outside the geographic coverage area **110** of a base station **102** or be otherwise unable to receive transmissions from a base station **102**. In some cases, groups of SL-UEs communicating via sidelink communications may utilize a one-to-many (1:M) system in which each SL-UE transmits to every other SL-UE in the group. In some cases, a base station **102** facilitates the scheduling of resources for sidelink communications. In other cases, sidelink communications are carried out between SL-UEs without the involvement of a base station **102**.

[0065] In an aspect, the sidelink **160** may operate over a wireless communication medium of interest, which may be shared with other wireless communications between other vehicles and/or infrastructure access points, as well as other RATs. A “medium” may be composed of one or more time, frequency, and/or space communication resources (e.g., encompassing one or more channels across one or more carriers) associated with wireless communication between one or more transmitter/receiver pairs. In an aspect, the medium of interest may correspond to at least a portion of an unlicensed frequency band shared among various RATs. Although different licensed frequency bands have been reserved for certain communication systems (e.g., by a government entity such as the Federal Communications Commission (FCC) in the United States), these systems, in particular those employing small cell access points, have recently extended operation into unlicensed frequency bands such as the Unlicensed National Information Infrastructure (U-NII) band used by wireless local area network (WLAN) technologies, most notably IEEE 802.11x WLAN technologies generally referred to as “Wi-Fi.” Example systems of this type include different variants of CDMA systems, TDMA systems, FDMA systems, orthogonal FDMA (OFDMA) systems, single-carrier FDMA (SC-FDMA) systems, and so on.

[0066] Note that although FIG. **1** only illustrates two of the UEs as SL-UEs (i.e., UEs **164** and **182**), any of the illustrated UEs may be SL-UEs. Further, although only UE **182** was described as being capable of beamforming, any of the illustrated UEs, including UE **164**, may be capable of beamforming. Where SL-UEs are capable of beamforming, they may beamform towards each other (i.e., towards other SL-UEs), towards other UEs (e.g., UEs **104**), towards base stations (e.g., base stations **102**, **180**, small cell **102'**, access point **150**), etc. Thus, in some cases, UEs **164** and **182** may utilize beamforming over sidelink **160**.

[0067] In the example of FIG. **1**, any of the illustrated UEs (shown in FIG. **1** as a single UE **104** for simplicity) may receive signals **124** from one or more Earth orbiting space vehicles (SVs) **112** (e.g., satellites). In an aspect, the SVs **112** may be part of a satellite positioning system that a UE **104** can use as an independent source of location information. A satellite positioning system typically includes a system of transmitters (e.g., SVs **112**) positioned to enable receivers (e.g., UEs **104**) to determine their location on or above the Earth based, at least in part, on positioning signals (e.g., signals **124**) received from the transmitters. Such a transmitter typically transmits a signal marked with a repeating pseudo-random noise (PN) code of a set number of chips. While typically located in SVs **112**, transmitters may sometimes be located on ground-based control stations, base stations **102**, and/or other UEs **104**. A UE **104** may include one or more dedicated receivers specifically designed to receive signals **124** for deriving geo location information from the SVs **112**.

[0068] In a satellite positioning system, the use of signals **124** can be augmented by various satellite-based augmentation systems (SBAS) that may be associated with or otherwise enabled for

use with one or more global and/or regional navigation satellite systems. For example an SBAS may include an augmentation system(s) that provides integrity information, differential corrections, etc., such as the Wide Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), the Multi-functional Satellite Augmentation System (MSAS), the Global Positioning System (GPS) Aided Geo Augmented Navigation or GPS and Geo Augmented Navigation system (GAGAN), and/or the like. Thus, as used herein, a satellite positioning system may include any combination of one or more global and/or regional navigation satellites associated with such one or more satellite positioning systems.

[0069] In an aspect, SVs **112** may additionally or alternatively be part of one or more non-terrestrial networks (NTNs). In an NTN, an SV **112** is connected to an earth station (also referred to as a ground station, NTN gateway, or gateway), which in turn is connected to an element in a 5G network, such as a modified base station **102** (without a terrestrial antenna) or a network node in a 5GC. This element would in turn provide access to other elements in the 5G network and ultimately to entities external to the 5G network, such as Internet web servers and other user devices. In that way, a UE **104** may receive communication signals (e.g., signals **124**) from an SV **112** instead of, or in addition to, communication signals from a terrestrial base station **102**.

[0070] The wireless communications system **100** may further include one or more UEs, such as UE **190**, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as “sidelinks”). In the example of FIG. **1**. UE **190** has a D2D P2P link **192** with one of the UEs **104** connected to one of the base stations **102** (e.g., through which UE **190** may indirectly obtain cellular connectivity) and a D2D P2P link **194** with WLAN STA **152** connected to the WLAN AP **150** (through which UE **190** may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links **192** and **194** may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), WiFi Direct (WiFi-D), Bluetooth®, and so on.

[0071] FIG. **2A** illustrates an example wireless network structure **200**. For example, a 5GC **210** (also referred to as a Next Generation Core (NGC)) can be viewed functionally as control plane (C-plane) functions **214** (e.g., UE registration, authentication, network access, gateway selection, etc.) and user plane (U-plane) functions **212**, (e.g., UE gateway function, access to data networks, IP routing, etc.) which operate cooperatively to form the core network. User plane interface (NG-U) **213** and control plane interface (NG-C) **215** connect the gNB **222** to the 5GC **210** and specifically to the user plane functions **212** and control plane functions **214**, respectively. In an additional configuration, an ng-eNB **224** may also be connected to the 5GC **210** via NG-C **215** to the control plane functions **214** and NG-U **213** to user plane functions **212**. Further, ng-eNB **224** may directly communicate with gNB **222** via a backhaul connection **223**. In some configurations, a Next Generation RAN (NG-RAN) **220** may have one or more gNBs **222**, while other configurations include one or more of both ng-eNBs **224** and gNBs **222**. Either (or both) gNB **222** or ng-eNB **224** may communicate with one or more UEs **204** (e.g., any of the UEs described herein).

[0072] Another optional aspect may include a location server **230**, which may be in communication with the 5GC **210** to provide location assistance for UE(s) **204**. The location server **230** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The location server **230** can be configured to support one or more location services for UEs **204** that can connect to the location server **230** via the core network. 5GC **210**, and/or via the Internet (not illustrated). Further, the location server **230** may be integrated into a component of the core network, or alternatively may be external to the core network (e.g., a third party server, such as an original equipment manufacturer (OEM) server or service server).

[0073] FIG. **2B** illustrates another example wireless network structure **250**. A 5GC **260** (which may correspond to 5GC **210** in FIG. **2A**) can be viewed functionally as control plane functions,

provided by an access and mobility management function (AMF) **264**, and user plane functions, provided by a user plane function (UPF) **262**, which operate cooperatively to form the core network (i.e., 5GC **260**). The functions of the AMF **264** include registration management, connection management, reachability management, mobility management, lawful interception, transport for session management (SM) messages between one or more UEs **204** (e.g., any of the UEs described herein) and a session management function (SMF) **266**, transparent proxy services for routing SM messages, access authentication and access authorization, transport for short message service (SMS) messages between the UE **204** and the short message service function (SMSF) (not shown), and security anchor functionality (SEAF). The AMF **264** also interacts with an authentication server function (AUSF) (not shown) and the UE **204**, and receives the intermediate key that was established as a result of the UE **204** authentication process. In the case of authentication based on a UMTS (universal mobile telecommunications system) subscriber identity module (USIM), the AMF **264** retrieves the security material from the AUSF. The functions of the AMF **264** also include security context management (SCM). The SCM receives a key from the SEAF that it uses to derive access-network specific keys. The functionality of the AMF **264** also includes location services management for regulatory services, transport for location services messages between the UE **204** and a location management function (LMF) **270** (which acts as a location server **230**), transport for location services messages between the NG-RAN **220** and the LMF **270**, evolved packet system (EPS) bearer identifier allocation for interworking with the EPS, and UE **204** mobility event notification. In addition, the AMF **264** also supports functionalities for non-3GPP (Third Generation Partnership Project) access networks.

[0074] Functions of the UPF **262** include acting as an anchor point for intra-/inter-RAT mobility (when applicable), acting as an external protocol data unit (PDU) session point of interconnect to a data network (not shown), providing packet routing and forwarding, packet inspection, user plane policy rule enforcement (e.g., gating, redirection, traffic steering), lawful interception (user plane collection), traffic usage reporting, quality of service (QoS) handling for the user plane (e.g., uplink/downlink rate enforcement, reflective QoS marking in the downlink), uplink traffic verification (service data flow (SDF) to QoS flow mapping), transport level packet marking in the uplink and downlink, downlink packet buffering and downlink data notification triggering, and sending and forwarding of one or more “end markers” to the source RAN node. The UPF **262** may also support transfer of location services messages over a user plane between the UE **204** and a location server, such as an SLP **272**.

[0075] The functions of the SMF **266** include session management, UE Internet protocol (IP) address allocation and management, selection and control of user plane functions, configuration of traffic steering at the UPF **262** to route traffic to the proper destination, control of part of policy enforcement and QoS, and downlink data notification. The interface over which the SMF **266** communicates with the AMF **264** is referred to as the N11 interface.

[0076] Another optional aspect may include an LMF **270**, which may be in communication with the 5GC **260** to provide location assistance for UEs **204**. The LMF **270** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The LMF **270** can be configured to support one or more location services for UEs **204** that can connect to the LMF **270** via the core network, 5GC **260**, and/or via the Internet (not illustrated). The SLP **272** may support similar functions to the LMF **270**, but whereas the LMF **270** may communicate with the AMF **264**, NG-RAN **220**, and UEs **204** over a control plane (e.g., using interfaces and protocols intended to convey signaling messages and not voice or data), the SLP **272** may communicate with UEs **204** and external clients (e.g., third-party server **274**) over a user plane (e.g., using protocols intended to carry voice and/or data like the transmission control protocol (TCP) and/or IP).

[0077] Yet another optional aspect may include a third-party server **274**, which may be in

communication with the LMF 270, the SLP 272, the 5GC 260 (e.g., via the AMF 264 and/or the UPF 262), the NG-RAN 220, and/or the UE 204 to obtain location information (e.g., a location estimate) for the UE 204. As such, in some cases, the third-party server 274 may be referred to as a location services (LCS) client or an external client. The third-party server 274 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server.

[0078] User plane interface 263 and control plane interface 265 connect the 5GC 260, and specifically the UPF 262 and AMF 264, respectively, to one or more gNBs 222 and/or ng-eNBs 224 in the NG-RAN 220. The interface between gNB(s) 222 and/or ng-eNB(s) 224 and the AMF 264 is referred to as the “N2” interface, and the interface between gNB(s) 222 and/or ng-eNB(s) 224 and the UPF 262 is referred to as the “N3” interface. The gNB(s) 222 and/or ng-eNB(s) 224 of the NG-RAN 220 may communicate directly with each other via backhaul connections 223, referred to as the “Xn-C” interface. One or more of gNBs 222 and/or ng-eNBs 224 may communicate with one or more UEs 204 over a wireless interface, referred to as the “Uu” interface. [0079] The functionality of a gNB 222 may be divided between a gNB central unit (gNB-CU) 226, one or more gNB distributed units (gNB-DUs) 228, and one or more gNB radio units (gNB-RUs) 229. A gNB-CU 226 is a logical node that includes the base station functions of transferring user data, mobility control, radio access network sharing, positioning, session management, and the like, except for those functions allocated exclusively to the gNB-DU(s) 228. More specifically, the gNB-CU 226 generally host the radio resource control (RRC), service data adaptation protocol (SDAP), and packet data convergence protocol (PDCP) protocols of the gNB 222. A gNB-DU 228 is a logical node that generally hosts the radio link control (RLC) and medium access control (MAC) layer of the gNB 222. Its operation is controlled by the gNB-CU 226. One gNB-DU 228 can support one or more cells, and one cell is supported by only one gNB-DU 228. The interface 232 between the gNB-CU 226 and the one or more gNB-DUs 228 is referred to as the “F1” interface. The physical (PHY) layer functionality of a gNB 222 is generally hosted by one or more standalone gNB-RUs 229 that perform functions such as power amplification and signal transmission/reception. The interface between a gNB-DU 228 and a gNB-RU 229 is referred to as the “Fx” interface. Thus, a UE 204 communicates with the gNB-CU 226 via the RRC, SDAP, and PDCP layers, with a gNB-DU 228 via the RLC and MAC layers, and with a gNB-RU 229 via the PHY layer.

[0080] FIGS. 3A, 3B, and 3C illustrate several example components (represented by corresponding blocks) that may be incorporated into a UE 302 (which may correspond to any of the UEs described herein), a base station 304 (which may correspond to any of the base stations described herein), and a network entity 306 (which may correspond to or embody any of the network functions described herein, including the location server 230 and the LMF 270, or alternatively may be independent from the NG-RAN 220 and/or 5GC 210/260 infrastructure depicted in FIGS. 2A and 2B, such as a private network) to support the file transmission operations as taught herein. It will be appreciated that these components may be implemented in different types of apparatuses in different implementations (e.g., in an ASIC, in a system-on-chip (SoC), etc.). The illustrated components may also be incorporated into other apparatuses in a communication system. For example, other apparatuses in a system may include components similar to those described to provide similar functionality. Also, a given apparatus may contain one or more of the components. For example, an apparatus may include multiple transceiver components that enable the apparatus to operate on multiple carriers and/or communicate via different technologies.

[0081] The UE 302 and the base station 304 each include one or more wireless wide area network (WWAN) transceivers 310 and 350, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) via one or more wireless communication networks (not shown), such as an

NR network, an LTE network, a GSM network, and/or the like. The WWAN transceivers **310** and **350** may each be connected to one or more antennas **316** and **356**, respectively, for communicating with other network nodes, such as other UEs, access points, base stations (e.g., eNBs, gNBs), etc., via at least one designated RAT (e.g., NR, LTE, GSM, etc.) over a wireless communication medium of interest (e.g., some set of time/frequency resources in a particular frequency spectrum). The WWAN transceivers **310** and **350** may be variously configured for transmitting and encoding signals **318** and **358** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **318** and **358** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the WWAN transceivers **310** and **350** include one or more transmitters **314** and **354**, respectively, for transmitting and encoding signals **318** and **358**, respectively, and one or more receivers **312** and **352**, respectively, for receiving and decoding signals **318** and **358**, respectively.

[0082] The UE **302** and the base station **304** each also include, at least in some cases, one or more short-range wireless transceivers **320** and **360**, respectively. The short-range wireless transceivers **320** and **360** may be connected to one or more antennas **326** and **366**, respectively, and provide means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) with other network nodes, such as other UEs, access points, base stations, etc., via at least one designated RAT (e.g., WiFi, LTE-D, Bluetooth®, Zigbee®, Z-Wave®, PC5, dedicated short-range communications (DSRC), wireless access for vehicular environments (WAVE), near-field communication (NFC), etc.) over a wireless communication medium of interest. The short-range wireless transceivers **320** and **360** may be variously configured for transmitting and encoding signals **328** and **368** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **328** and **368** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the short-range wireless transceivers **320** and **360** include one or more transmitters **324** and **364**, respectively, for transmitting and encoding signals **328** and **368**, respectively, and one or more receivers **322** and **362**, respectively, for receiving and decoding signals **328** and **368**, respectively. As specific examples, the short-range wireless transceivers **320** and **360** may be WiFi transceivers, Bluetooth® transceivers, Zigbee® and/or Z-Wave® transceivers, NFC transceivers, or vehicle-to-vehicle (V2V) and/or vehicle-to-everything (V2X) transceivers.

[0083] The UE **302** and the base station **304** also include, at least in some cases, satellite signal receivers **330** and **370**. The satellite signal receivers **330** and **370** may be connected to one or more antennas **336** and **376**, respectively, and may provide means for receiving and/or measuring satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal receivers **330** and **370** are satellite positioning system receivers, the satellite positioning/communication signals **338** and **378** may be global positioning system (GPS) signals, global navigation satellite system (GLONASS) signals, Galileo signals, Beidou signals, Indian Regional Navigation Satellite System (NAVIC), Quasi-Zenith Satellite System (QZSS), etc. Where the satellite signal receivers **330** and **370** are non-terrestrial network (NTN) receivers, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal receivers **330** and **370** may comprise any suitable hardware and/or software for receiving and processing satellite positioning/communication signals **338** and **378**, respectively. The satellite signal receivers **330** and **370** may request information and operations as appropriate from the other systems, and, at least in some cases, perform calculations to determine locations of the UE **302** and the base station **304**, respectively, using measurements obtained by any suitable satellite positioning system algorithm.

[0084] The base station **304** and the network entity **306** each include one or more network transceivers **380** and **390**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, etc.) with other network entities (e.g., other base stations **304**,

other network entities **306**). For example, the base station **304** may employ the one or more network transceivers **380** to communicate with other base stations **304** or network entities **306** over one or more wired or wireless backhaul links. As another example, the network entity **306** may employ the one or more network transceivers **390** to communicate with one or more base station **304** over one or more wired or wireless backhaul links, or with other network entities **306** over one or more wired or wireless core network interfaces.

[0085] A transceiver may be configured to communicate over a wired or wireless link. A transceiver (whether a wired transceiver or a wireless transceiver) includes transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) and receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**). A transceiver may be an integrated device (e.g., embodying transmitter circuitry and receiver circuitry in a single device) in some implementations, may comprise separate transmitter circuitry and separate receiver circuitry in some implementations, or may be embodied in other ways in other implementations. The transmitter circuitry and receiver circuitry of a wired transceiver (e.g., network transceivers **380** and **390** in some implementations) may be coupled to one or more wired network interface ports. Wireless transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform transmit “beamforming,” as described herein. Similarly, wireless receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform receive beamforming, as described herein. In an aspect, the transmitter circuitry and receiver circuitry may share the same plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such that the respective apparatus can only receive or transmit at a given time, not both at the same time. A wireless transceiver (e.g., WWAN transceivers **310** and **350**, short-range wireless transceivers **320** and **360**) may also include a network listen module (NLM) or the like for performing various measurements.

[0086] As used herein, the various wireless transceivers (e.g., transceivers **310**, **320**, **350**, and **360**, and network transceivers **380** and **390** in some implementations) and wired transceivers (e.g., network transceivers **380** and **390** in some implementations) may generally be characterized as “a transceiver,” “at least one transceiver,” or “one or more transceivers.” As such, whether a particular transceiver is a wired or wireless transceiver may be inferred from the type of communication performed. For example, backhaul communication between network devices or servers will generally relate to signaling via a wired transceiver, whereas wireless communication between a UE (e.g., UE **302**) and a base station (e.g., base station **304**) will generally relate to signaling via a wireless transceiver.

[0087] The UE **302**, the base station **304**, and the network entity **306** also include other components that may be used in conjunction with the operations as disclosed herein. The UE **302**, the base station **304**, and the network entity **306** include one or more processors **332**, **384**, and **394**, respectively, for providing functionality relating to, for example, wireless communication, and for providing other processing functionality. The processors **332**, **384**, and **394** may therefore provide means for processing, such as means for determining, means for calculating, means for receiving, means for transmitting, means for indicating, etc. In an aspect, the processors **332**, **384**, and **394** may include, for example, one or more general purpose processors, multi-core processors, central processing units (CPUs), ASICs, digital signal processors (DSPs), field programmable gate arrays (FPGAs), other programmable logic devices or processing circuitry, or various combinations thereof.

[0088] The UE **302**, the base station **304**, and the network entity **306** include memory circuitry implementing memories **340**, **386**, and **396** (e.g., each including a memory device), respectively, for maintaining information (e.g., information indicative of reserved resources, thresholds, parameters, and so on). The memories **340**, **386**, and **396** may therefore provide means for storing,

means for retrieving, means for maintaining, etc. In some cases, the UE **302**, the base station **304**, and the network entity **306** may include PRS component **342**, **388**, and **398**, respectively. The PRS component **342**, **388**, and **398** may be hardware circuits that are part of or coupled to the processors **332**, **384**, and **394**, respectively, that, when executed, cause the UE **302**, the base station **304**, and the network entity **306** to perform the functionality described herein. In other aspects, the PRS component **342**, **388**, and **398** may be external to the processors **332**, **384**, and **394** (e.g., part of a modem processing system, integrated with another processing system, etc.). Alternatively, the PRS component **342**, **388**, and **398** may be memory modules stored in the memories **340**, **386**, and **396**, respectively, that, when executed by the processors **332**, **384**, and **394** (or a modem processing system, another processing system, etc.), cause the UE **302**, the base station **304**, and the network entity **306** to perform the functionality described herein. FIG. 3A illustrates possible locations of the PRS component **342**, which may be, for example, part of the one or more WWAN transceivers **310**, the memory **340**, the one or more processors **332**, or any combination thereof, or may be a standalone component. FIG. 3B illustrates possible locations of the PRS component **388**, which may be, for example, part of the one or more WWAN transceivers **350**, the memory **386**, the one or more processors **384**, or any combination thereof, or may be a standalone component. FIG. 3C illustrates possible locations of the PRS component **398**, which may be, for example, part of the one or more network transceivers **390**, the memory **396**, the one or more processors **394**, or any combination thereof, or may be a standalone component.

[0089] The UE **302** may include one or more sensors **344** coupled to the one or more processors **332** to provide means for sensing or detecting movement and/or orientation information that is independent of motion data derived from signals received by the one or more WWAN transceivers **310**, the one or more short-range wireless transceivers **320**, and/or the satellite signal receiver **330**. By way of example, the sensor(s) **344** may include an accelerometer (e.g., a micro-electrical mechanical systems (MEMS) device), a gyroscope, a geomagnetic sensor (e.g., a compass), an altimeter (e.g., a barometric pressure altimeter), and/or any other type of movement detection sensor. Moreover, the sensor(s) **344** may include a plurality of different types of devices and combine their outputs in order to provide motion information. For example, the sensor(s) **344** may use a combination of a multi-axis accelerometer and orientation sensors to provide the ability to compute positions in two-dimensional (2D) and/or three-dimensional (3D) coordinate systems. [0090] In addition, the UE **302** includes a user interface **346** providing means for providing indications (e.g., audible and/or visual indications) to a user and/or for receiving user input (e.g., upon user actuation of a sensing device such a keypad, a touch screen, a microphone, and so on). Although not shown, the base station **304** and the network entity **306** may also include user interfaces.

[0091] Referring to the one or more processors **384** in more detail, in the downlink, IP packets from the network entity **306** may be provided to the processor **384**. The one or more processors **384** may implement functionality for an RRC layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The one or more processors **384** may provide RRC layer functionality associated with broadcasting of system information (e.g., master information block (MIB), system information blocks (SIBs)), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter-RAT mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer PDUs, error correction through automatic repeat request (ARQ), concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, scheduling information reporting, error

correction, priority handling, and logical channel prioritization.

[0092] The transmitter **354** and the receiver **352** may implement Layer-1 (L1) functionality associated with various signal processing functions. Layer-1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The transmitter **354** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an orthogonal frequency division multiplexing (OFDM) subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an inverse fast Fourier transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM symbol stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **302**. Each spatial stream may then be provided to one or more different antennas **356**. The transmitter **354** may modulate an RF carrier with a respective spatial stream for transmission.

[0093] At the UE **302**, the receiver **312** receives a signal through its respective antenna(s) **316**. The receiver **312** recovers information modulated onto an RF carrier and provides the information to the one or more processors **332**. The transmitter **314** and the receiver **312** implement Layer-1 functionality associated with various signal processing functions. The receiver **312** may perform spatial processing on the information to recover any spatial streams destined for the UE **302**. If multiple spatial streams are destined for the UE **302**, they may be combined by the receiver **312** into a single OFDM symbol stream. The receiver **312** then converts the OFDM symbol stream from the time-domain to the frequency domain using a fast Fourier transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **304**. These soft decisions may be based on channel estimates computed by a channel estimator. The soft decisions are then decoded and de-interleaved to recover the data and control signals that were originally transmitted by the base station **304** on the physical channel. The data and control signals are then provided to the one or more processors **332**, which implements Layer-3 (L3) and Layer-2 (L2) functionality.

[0094] In the uplink, the one or more processors **332** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the core network. The one or more processors **332** are also responsible for error detection.

[0095] Similar to the functionality described in connection with the downlink transmission by the base station **304**, the one or more processors **332** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through hybrid automatic repeat request (HARQ), priority handling, and logical channel

prioritization.

[0096] Channel estimates derived by the channel estimator from a reference signal or feedback transmitted by the base station **304** may be used by the transmitter **314** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the transmitter **314** may be provided to different antenna(s) **316**. The transmitter **314** may modulate an RF carrier with a respective spatial stream for transmission.

[0097] The uplink transmission is processed at the base station **304** in a manner similar to that described in connection with the receiver function at the UE **302**. The receiver **352** receives a signal through its respective antenna(s) **356**. The receiver **352** recovers information modulated onto an RF carrier and provides the information to the one or more processors **384**.

[0098] In the uplink, the one or more processors **384** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE **302**. IP packets from the one or more processors **384** may be provided to the core network. The one or more processors **384** are also responsible for error detection.

[0099] For convenience, the UE **302**, the base station **304**, and/or the network entity **306** are shown in FIGS. **3A**, **3B**, and **3C** as including various components that may be configured according to the various examples described herein. It will be appreciated, however, that the illustrated components may have different functionality in different designs. In particular, various components in FIGS. **3A** to **3C** are optional in alternative configurations and the various aspects include configurations that may vary due to design choice, costs, use of the device, or other considerations. For example, in case of FIG. **3A**, a particular implementation of UE **302** may omit the WWAN transceiver(s) **310** (e.g., a wearable device or tablet computer or PC or laptop may have Wi-Fi and/or Bluetooth capability without cellular capability), or may omit the short-range wireless transceiver(s) **320** (e.g., cellular-only, etc.), or may omit the satellite signal receiver **330**, or may omit the sensor(s) **344**, and so on. In another example, in case of FIG. **3B**, a particular implementation of the base station **304** may omit the WWAN transceiver(s) **350** (e.g., a Wi-Fi “hotspot” access point without cellular capability), or may omit the short-range wireless transceiver(s) **360** (e.g., cellular-only, etc.), or may omit the satellite receiver **370**, and so on. For brevity, illustration of the various alternative configurations is not provided herein, but would be readily understandable to one skilled in the art.

[0100] The various components of the UE **302**, the base station **304**, and the network entity **306** may be communicatively coupled to each other over data buses **334**, **382**, and **392**, respectively. In an aspect, the data buses **334**, **382**, and **392** may form, or be part of, a communication interface of the UE **302**, the base station **304**, and the network entity **306**, respectively. For example, where different logical entities are embodied in the same device (e.g., gNB and location server functionality incorporated into the same base station **304**), the data buses **334**, **382**, and **392** may provide communication between them.

[0101] The components of FIGS. **3A**, **3B**, and **3C** may be implemented in various ways. In some implementations, the components of FIGS. **3A**, **3B**, and **3C** may be implemented in one or more circuits such as, for example, one or more processors and/or one or more ASICs (which may include one or more processors). Here, each circuit may use and/or incorporate at least one memory component for storing information or executable code used by the circuit to provide this functionality. For example, some or all of the functionality represented by blocks **310** to **346** may be implemented by processor and memory component(s) of the UE **302** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Similarly, some or all of the functionality represented by blocks **350** to **388** may be implemented by processor and memory component(s) of the base station **304** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Also, some or all of the functionality represented by blocks **390** to **398** may be implemented by processor and memory component(s) of the network entity **306** (e.g., by execution of appropriate code and/or by appropriate configuration

of processor components). For simplicity, various operations, acts, and/or functions are described herein as being performed “by a UE,” “by a base station,” “by a network entity,” etc. However, as will be appreciated, such operations, acts, and/or functions may actually be performed by specific components or combinations of components of the UE **302**, base station **304**, network entity **306**, etc., such as the processors **332**, **384**, **394**, the transceivers **310**, **320**, **350**, and **360**, the memories **340**, **386**, and **396**, the PRS component **342**, **388**, and **398**, etc.

[0102] In some designs, the network entity **306** may be implemented as a core network component. In other designs, the network entity **306** may be distinct from a network operator or operation of the cellular network infrastructure (e.g., NG RAN **220** and/or 5GC **210/260**). For example, the network entity **306** may be a component of a private network that may be configured to communicate with the UE **302** via the base station **304** or independently from the base station **304** (e.g., over a non-cellular communication link, such as WiFi).

[0103] Note that the UE **302** illustrated in FIG. **3A** may represent a “low-tier” UE or a “premium” UE. As described further below, while low-tier and premium UEs may have the same types of components (e.g., both may have WWAN transceivers **310**, processing systems **332**, memory components **340**, etc.), the components may have different degrees of functionality (e.g., increased or decreased performance, more or fewer capabilities, etc.) depending on whether the UE **302** corresponds to a low-tier UE or a premium UE.

[0104] Various frame structures may be used to support downlink and uplink transmissions between network nodes (e.g., base stations and UEs). FIG. **4** is a diagram **400** illustrating an example frame structure, according to aspects of the disclosure. The frame structure may be a downlink or uplink frame structure. Other wireless communications technologies may have different frame structures and/or different channels.

[0105] LTE, and in some cases NR, utilizes OFDM on the downlink and single-carrier frequency division multiplexing (SC-FDM) on the uplink. Unlike LTE, however, NR has an option to use OFDM on the uplink as well. OFDM and SC-FDM partition the system bandwidth into multiple (K) orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with data. In general, modulation symbols are sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system bandwidth. For example, the spacing of the subcarriers may be 15 kilohertz (kHz) and the minimum resource allocation (resource block) may be 12 subcarriers (or 180 kHz). Consequently, the nominal FFT size may be equal to 128, 256, 512, 1024, or 2048 for system bandwidth of 1.25, 2.5, 5, 10, or 20 megahertz (MHz), respectively. The system bandwidth may also be partitioned into subbands. For example, a subband may cover 1.08 MHz (i.e., 6 resource blocks), and there may be 1, 2, 4, 8, or 16 subbands for system bandwidth of 1.25, 2.5, 5, 10, or 20 MHz, respectively.

[0106] LTE supports a single numerology (subcarrier spacing (SCS), symbol length, etc.). In contrast, NR may support multiple numerologies (μ), for example, subcarrier spacings of 15 kHz ($\mu=0$), 30 kHz ($\mu=1$), 60 kHz ($\mu=2$), 120 kHz ($\mu=3$), and 240 kHz ($\mu=4$) or greater may be available. In each subcarrier spacing, there are 14 symbols per slot. For 15 kHz SCS ($\mu=0$), there is one slot per subframe, 10 slots per frame, the slot duration is 1 millisecond (ms), the symbol duration is 66.7 microseconds (μ s), and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 50. For 30 kHz SCS ($\mu=1$), there are two slots per subframe, 20 slots per frame, the slot duration is 0.5 ms, the symbol duration is 33.3 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 100. For 60 kHz SCS ($\mu=2$), there are four slots per subframe, 40 slots per frame, the slot duration is 0.25 ms, the symbol duration is 16.7 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 200. For 120 kHz SCS ($\mu=3$), there are eight slots per subframe, 80 slots per frame, the slot duration is 0.125 ms, the symbol duration is 8.33 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 400. For 240 kHz SCS ($\mu=4$), there are 16 slots per subframe, 160 slots per frame, the slot

duration is 0.0625 ms, the symbol duration is 4.17 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 800.

[0107] In the example of FIG. 4, a numerology of 15 kHz is used. Thus, in the time domain, a 10 ms frame is divided into 10 equally sized subframes of 1 ms each, and each subframe includes one time slot. In FIG. 4, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top.

[0108] A resource grid may be used to represent time slots, each time slot including one or more time-concurrent resource blocks (RBs) (also referred to as physical RBs (PRBs)) in the frequency domain. The resource grid is further divided into multiple resource elements (REs). An RE may correspond to one symbol length in the time domain and one subcarrier in the frequency domain. In the numerology of FIG. 4, for a normal cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and seven consecutive symbols in the time domain, for a total of 84 REs. For an extended cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and six consecutive symbols in the time domain, for a total of 72 REs. The number of bits carried by each RE depends on the modulation scheme.

[0109] Some of the REs may carry reference (pilot) signals (RS). The reference signals may include positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signals (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), sounding reference signals (SRS), etc., depending on whether the illustrated frame structure is used for uplink or downlink communication. FIG. 4 illustrates example locations of REs carrying a reference signal (labeled “R”).

[0110] FIG. 5 is a diagram 500 illustrating various downlink channels within an example downlink slot. In FIG. 5, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top. In the example of FIG. 5, a numerology of 15 kHz is used. Thus, in the time domain, the illustrated slot is one millisecond (ms) in length, divided into 14 symbols.

[0111] In NR, the channel bandwidth, or system bandwidth, is divided into multiple bandwidth parts (BWPs). A BWP is a contiguous set of RBs selected from a contiguous subset of the common RBs for a given numerology on a given carrier. Generally, a maximum of four BWPs can be specified in the downlink and uplink. That is, a UE can be configured with up to four BWPs on the downlink, and up to four BWPs on the uplink. Only one BWP (uplink or downlink) may be active at a given time, meaning the UE may only receive or transmit over one BWP at a time. On the downlink, the bandwidth of each BWP should be equal to or greater than the bandwidth of the SSB, but it may or may not contain the SSB.

[0112] Referring to FIG. 5, a primary synchronization signal (PSS) is used by a UE to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a PCI. Based on the PCI, the UE can determine the locations of the aforementioned DL-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form an SSB (also referred to as an SS/PBCH). The MIB provides a number of RBs in the downlink system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH, such as system information blocks (SIBs), and paging messages.

[0113] The physical downlink control channel (PDCCH) carries downlink control information (DCI) within one or more control channel elements (CCEs), each CCE including one or more RE

group (REG) bundles (which may span multiple symbols in the time domain), each REG bundle including one or more REGs, each REG corresponding to 12 resource elements (one resource block) in the frequency domain and one OFDM symbol in the time domain. The set of physical resources used to carry the PDCCH/DCI is referred to in NR as the control resource set (CORESET). In NR, a PDCCH is confined to a single CORESET and is transmitted with its own DMRS. This enables UE-specific beamforming for the PDCCH.

[0114] In the example of FIG. 5, there is one CORESET per BWP, and the CORESET spans three symbols (although it may be only one or two symbols) in the time domain. Unlike LTE control channels, which occupy the entire system bandwidth, in NR, PDCCH channels are localized to a specific region in the frequency domain (i.e., a CORESET). Thus, the frequency component of the PDCCH shown in FIG. 5 is illustrated as less than a single BWP in the frequency domain. Note that although the illustrated CORESET is contiguous in the frequency domain, it need not be. In addition, the CORESET may span less than three symbols in the time domain.

[0115] The DCI within the PDCCH carries information about uplink resource allocation (persistent and non-persistent) and descriptions about downlink data transmitted to the UE, referred to as uplink and downlink grants, respectively. More specifically, the DCI indicates the resources scheduled for the downlink data channel (e.g., PDSCH) and the uplink data channel (e.g., physical uplink shared channel (PUSCH)). Multiple (e.g., up to eight) DCIs can be configured in the PDCCH, and these DCIs can have one of multiple formats. For example, there are different DCI formats for uplink scheduling, for downlink scheduling, for uplink transmit power control (TPC), etc. A PDCCH may be transported by 1, 2, 4, 8, or 16 CCEs in order to accommodate different DCI payload sizes or coding rates.

[0116] FIG. 6 is a diagram **600** illustrating various uplink channels within an example uplink slot. In FIG. 6, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top. In the example of FIG. 6, a numerology of 15 kHz is used. Thus, in the time domain, the illustrated slot is one millisecond (ms) in length, divided into 14 symbols.

[0117] A random-access channel (RACH), also referred to as a physical random-access channel (PRACH), may be within one or more slots within a frame based on the PRACH configuration. The PRACH may include six consecutive RB pairs within a slot. The PRACH allows the UE to perform initial system access and achieve uplink synchronization. A physical uplink control channel (PUCCH) may be located on edges of the uplink system bandwidth. The PUCCH carries uplink control information (UCI), such as scheduling requests, CSI reports, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and HARQ ACK/NACK feedback. The physical uplink shared channel (PUSCH) carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

[0118] FIG. 7 illustrates time and frequency resources used for sidelink communication. A time-frequency grid **700** is divided into subchannels in the frequency domain and is divided into time slots in the time domain. Each subchannel comprises a number (e.g., 10, 15, 20, 25, 50, 75, or 100) of physical resource blocks (PRBs), and each slot contains a number (e.g., 14) of OFDM symbols. A sidelink communication can be (pre) configured to occupy fewer than 14 symbols in a slot. The first symbol of the slot is repeated on the preceding symbol for automatic gain control (AGC) settling. The example slot shown in FIG. 4 contains a physical sidelink control channel (PSCCH) portion and a physical sidelink shared channel (PSSCH) portion, with a gap symbol following the PSCCH. PSCCH and PSSCH are transmitted in the same slot.

[0119] Sidelink communications take place within transmission or reception resource pools. Sidelink communications occupy one slot and one or more subchannels. Some slots are not available for sidelink, and some slots contain feedback resources. Sidelink communication can be preconfigured (e.g., preloaded on a UE) or configured (e.g., by a base station via RRC).

[0120] FIG. 8 is a diagram of an example PRS configuration **800** for the PRS transmissions of a

given base station, according to aspects of the disclosure. In FIG. 8, time is represented horizontally, increasing from left to right. Each long rectangle represents a slot and each short (shaded) rectangle represents an OFDM symbol. In the example of FIG. 8, a PRS resource set **810** (labeled “PRS resource set 1”) includes two PRS resources, a first PRS resource **812** (labeled “PRS resource 1”) and a second PRS resource **814** (labeled “PRS resource 2”). The base station transmits PRS on the PRS resources **812** and **814** of the PRS resource set **810**.

[0121] The PRS resource set **810** has an occasion length (N_{PRS}) of two slots and a periodicity (T_{PRS}) of, for example, 160 slots or 160 milliseconds (ms) (for 15 kHz subcarrier spacing). As such, both the PRS resources **812** and **814** are two consecutive slots in length and repeat every T_{PRS} slots, starting from the slot in which the first symbol of the respective PRS resource occurs. In the example of FIG. 8, the PRS resource **812** has a symbol length (N_{ymb}) of two symbols, and the PRS resource **814** has a symbol length (N_{ymb}) of four symbols. The PRS resource **812** and the PRS resource **814** may be transmitted on separate beams of the same base station.

[0122] Each instance of the PRS resource set **810**, illustrated as instances **820a**, **820b**, and **820c**, includes an occasion of length: ‘2’ (i.e., $N_{\text{PRS}}=2$) for each PRS resource **812**, **814** of the PRS resource set. The PRS resources **812** and **814** are repeated every T_{PRS} slots up to the muting sequence periodicity T_{REP} . As such, a bitmap of length T_{REP} would be needed to indicate which occasions of instances **820a**, **820b**, and **820c** of PRS resource set **810** are muted (i.e., not transmitted).

[0123] In an aspect, there may be additional constraints on the PRS configuration **800**. For example, for all PRS resources (e.g., PRS resources **812**, **814**) of a PRS resource set (e.g., PRS resource set **810**), the base station can configure the following parameters to be the same: (a) the occasion length (N_{PRS}). (b) the number of symbols (N_{ymb}), (c) the comb type, and/or (d) the bandwidth. In addition, for all PRS resources of all PRS resource sets, the subcarrier spacing and the cyclic prefix can be configured to be the same for one base station or for all base stations. Whether it is for one base station or all base stations may depend on the UE's capability to support the first and/or second option.

[0124] FIG. 9 is a diagram **900** illustrating an example PRS configuration for two TRPs (labeled “TRP1” and “TRP2”) operating in the same positioning frequency layer (labeled “Positioning Frequency Layer 1”), according to aspects of the disclosure. For a positioning session, a UE may be provided with assistance data indicating the illustrated PRS configuration. In the example of FIG. 9, the first TRP (“TRP1”) is associated with (e.g., transmits) two PRS resource sets, labeled “PRS Resource Set 1” and “PRS Resource Set 2,” and the second TRP (“TRP2”) is associated with one PRS resource set, labeled “PRS Resource Set 3.” Each PRS resource set comprises at least two PRS resources. Specifically, the first PRS resource set (“PRS Resource Set 1”) includes PRS resources labeled “PRS Resource 1” and “PRS Resource 2,” the second PRS resource set (“PRS Resource Set 2”) includes PRS resources labeled “PRS Resource 3” and “PRS Resource 4,” and the third PRS resource set (“PRS Resource Set 3”) includes PRS resources labeled “PRS Resource 5” and “PRS Resource 6.”

[0125] FIG. 10 is a graph **1000** representing the channel impulse response of a multipath channel between a receiver device (e.g., any of the UEs or base stations described herein) and a transmitter device (e.g., any other of the UEs or base stations described herein), according to aspects of the disclosure. The channel impulse response represents the intensity of a radio frequency (RF) signal received through a multipath channel as a function of time delay. Thus, the horizontal axis is in units of time (e.g., milliseconds) and the vertical axis is in units of signal strength (e.g., decibels). Note that a multipath channel is a channel between a transmitter and a receiver over which an RF signal follows multiple paths, or multipaths, due to transmission of the RF signal on multiple beams and/or to the propagation characteristics of the RF signal (e.g., reflection, refraction, etc.).

[0126] In the example of FIG. 10, the receiver detects/measures multiple (four) clusters of channel taps. Each channel tap represents a multipath that an RF signal followed between the transmitter

and the receiver. That is, a channel tap represents the arrival of an RF signal on a multipath. Each cluster of channel taps indicates that the corresponding multipaths followed essentially the same path. There may be different clusters due to the RF signal being transmitted on different transmit beams (and therefore at different angles), or because of the propagation characteristics of RF signals (e.g., potentially following different paths due to reflections), or both.

[0127] All of the clusters of channel taps for a given RF signal represent the multipath channel (or simply channel) between the transmitter and receiver. Under the channel illustrated in FIG. 10, the receiver receives a first cluster of two RF signals on channel taps at time T1, a second cluster of five RF signals on channel taps at time T2, a third cluster of five RF signals on channel taps at time T3, and a fourth cluster of four RF signals on channel taps at time T4. In the example of FIG. 10, because the first cluster of RF signals at time T1 arrives first, it is assumed to correspond to the RF signal transmitted on the transmit beam aligned with the line-of-sight (LOS), or the shortest, path. The third cluster at time T3 is comprised of the strongest RF signals, and may correspond to, for example, the RF signal transmitted on a transmit beam aligned with a non-line-of-sight (NLOS) path. Note that although FIG. 10 illustrates clusters of two to five channel taps, as will be appreciated, the clusters may have more or fewer than the illustrated number of channel taps.

[0128] NR supports a number of cellular network-based positioning technologies, including downlink-based, uplink-based, and downlink-and-uplink-based positioning methods. Downlink-based positioning methods include observed time difference of arrival (OTDOA) in LTE, downlink time difference of arrival (DL-TDOA) in NR, and downlink angle-of-departure (DL-AoD) in NR. FIG. 11 illustrates examples of various positioning methods, according to aspects of the disclosure. In an OTDOA or DL-TDOA positioning procedure, illustrated by scenario 1110, a UE measures the differences between the times of arrival (ToAs) of reference signals (e.g., positioning reference signals (PRS)) received from pairs of base stations, referred to as reference signal time difference (RSTD) or time difference of arrival (TDOA) measurements, and reports them to a positioning entity. More specifically, the UE receives the identifiers (IDs) of a reference base station (e.g., a serving base station) and multiple non-reference base stations in assistance data. The UE then measures the RSTD between the reference base station and each of the non-reference base stations. Based on the known locations of the involved base stations and the RSTD measurements, the positioning entity (e.g., the UE for UE-based positioning or a location server for UE-assisted positioning) can estimate the UE's location.

[0129] For DL-AoD positioning, illustrated by scenario 1120, the positioning entity uses a measurement report from the UE of received signal strength measurements of multiple downlink transmit beams to determine the angle(s) between the UE and the transmitting base station(s). The positioning entity can then estimate the location of the UE based on the determined angle(s) and the known location(s) of the transmitting base station(s).

[0130] Uplink-based positioning methods include uplink time difference of arrival (UL-TDOA) and uplink angle-of-arrival (UL-AoA). UL-TDOA is similar to DL-TDOA, but is based on uplink reference signals (e.g., sounding reference signals (SRS)) transmitted by the UE. For UL-AoA positioning, one or more base stations measure the received signal strength of one or more uplink reference signals (e.g., SRS) received from a UE on one or more uplink receive beams. The positioning entity uses the signal strength measurements and the angle(s) of the receive beam(s) to determine the angle(s) between the UE and the base station(s). Based on the determined angle(s) and the known location(s) of the base station(s), the positioning entity can then estimate the location of the UE.

[0131] Downlink-and-uplink-based positioning methods include enhanced cell-ID (E-CID) positioning and multi-round-trip-time (RTT) positioning (also referred to as “multi-cell RTT” and “multi-RTT”). In an RTT procedure, a first entity (e.g., a base station or a UE) transmits a first RTT-related signal (e.g., a PRS or SRS) to a second entity (e.g., a UE or base station), which transmits a second RTT-related signal (e.g., an SRS or PRS) back to the first entity. Each entity

measures the time difference between the time of arrival (ToA) of the received RTT-related signal and the transmission time of the transmitted RTT-related signal. This time difference is referred to as a reception-to-transmission (Rx-Tx) time difference. The Rx-Tx time difference measurement may be made, or may be adjusted, to include only a time difference between nearest slot boundaries for the received and transmitted signals. Both entities may then send their Rx-Tx time difference measurement to a location server (e.g., an LMF **270**), which calculates the round trip propagation time (i.e., RTT) between the two entities from the two Rx-Tx time difference measurements (e.g., as the sum of the two Rx-Tx time difference measurements). Alternatively, one entity may send its Rx-Tx time difference measurement to the other entity, which then calculates the RTT. The distance between the two entities can be determined from the RTT and the known signal speed (e.g., the speed of light). For multi-RTT positioning, illustrated by scenario **1130**, a first entity (e.g., a UE or base station) performs an RTT positioning procedure with multiple second entities (e.g., multiple base stations or UEs) to enable the location of the first entity to be determined (e.g., using multilateration) based on distances to, and the known locations of, the second entities. RTT and multi-RTT methods can be combined with other positioning techniques, such as UL-AoA and DL-AoD, to improve location accuracy, as illustrated by scenario **1140**.

[0132] The E-CID positioning method is based on radio resource management (RRM) measurements. In E-CID, the UE reports the serving cell ID, the timing advance (TA), and the identifiers, estimated timing, and signal strength of detected neighbor base stations. The location of the UE is then estimated based on this information and the known locations of the base station(s).

[0133] To assist positioning operations, a location server (e.g., location server **230**, LMF **270**, SLP **272**) may provide assistance data to the UE. For example, the assistance data may include identifiers of the base stations (or the cells/TRPs of the base stations) from which to measure reference signals, the reference signal configuration parameters (e.g., the number of consecutive slots including PRS, periodicity of the consecutive slots including PRS, muting sequence, frequency hopping sequence, reference signal identifier, reference signal bandwidth, etc.), and/or other parameters applicable to the particular positioning method. Alternatively, the assistance data may originate directly from the base stations themselves (e.g., in periodically broadcasted overhead messages, etc.). In some cases, the UE may be able to detect neighbor network nodes itself without the use of assistance data.

[0134] In the case of an OTDOA or DL-TDOA positioning procedure, the assistance data may further include an expected RSTD value and an associated uncertainty, or search window, around the expected RSTD. In some cases, the value range of the expected RSTD may be ± 500 microseconds (μs). In some cases, when any of the resources used for the positioning measurement are in FR1, the value range for the uncertainty of the expected RSTD may be ± 32 μs . In other cases, when all of the resources used for the positioning measurement(s) are in FR2, the value range for the uncertainty of the expected RSTD may be ± 8 μs .

[0135] A location estimate may be referred to by other names, such as a position estimate, location, position, position fix, fix, or the like. A location estimate may be geodetic and comprise coordinates (e.g., latitude, longitude, and possibly altitude) or may be civic and comprise a street address, postal address, or some other verbal description of a location. A location estimate may further be defined relative to some other known location or defined in absolute terms (e.g., using latitude, longitude, and possibly altitude). A location estimate may include an expected error or uncertainty (e.g., by including an area or volume within which the location is expected to be included with some specified or default level of confidence).

[0136] In NR, there may not be precise timing synchronization across the network. Instead, it may be sufficient to have coarse time-synchronization across base stations (e.g., within a cyclic prefix (CP) duration of the orthogonal frequency division multiplexing (OFDM) symbols). RTT-based methods generally only need coarse timing synchronization, and as such, are a preferred positioning method in NR.

[0137] FIG. 12 illustrates an example wireless communications system 1200, according to aspects of the disclosure. In the example of FIG. 12, a UE 1204 (e.g., any of the UEs described herein) is attempting to calculate an estimate of its location, or assist another entity (e.g., a base station or core network component, another UE, a location server, a third party application, etc.) to calculate an estimate of its location. The UE 1204 may transmit and receive wireless signals to and from a plurality of network nodes (labeled “Node”) 1202-1, 1202-2, and 1202-3 (collectively, network nodes 1202). The network nodes 1202 may include one or more base stations (e.g., any of the base stations described herein), one or more reconfigurable intelligent displays (RIS), one or more positioning beacons, one or more UEs (e.g., connected over sidelinks), etc.

[0138] In a network-centric RTT positioning procedure the serving base station (e.g., one of network nodes 1202) instructs the UE 1204 to measure RTT measurement signals (e.g., PRS) from two or more neighboring network nodes 1202 (and typically the serving base station, as at least three network nodes 1202 are needed for a two-dimensional location estimate). The involved network nodes 1202 transmit RTT measurement signals on low reuse resources (e.g., resources used by the network nodes 1202 to transmit system information, where the network nodes 1202 are base stations) allocated by the network (e.g., location server 230, LMF 270, SLP 272). The UE 1204 records the arrival time (also referred to as the receive time, reception time, time of reception, or time of arrival) of each RTT measurement signal relative to the UE's 1204 current downlink timing (e.g., as derived by the UE 1204 from a downlink signal received from its serving base station), and transmits a common or individual RTT response signal (e.g., SRS) to the involved network nodes 1202 on resources allocated by its serving base station. The UE 1204, if it not the positioning entity, reports a UE reception-to-transmission (Rx-Tx) time difference measurement to the positioning entity. The UE Rx-Tx time difference measurement indicates the time difference between the arrival time of each RTT measurement signal at the UE 1204 and the transmission time(s) of the RTT response signal(s). Each involved network node 1202 also reports, to the positioning entity, a network node Rx-Tx time difference measurement (also referred to as a base station (BS) or gNB Rx-Tx time difference measurement), which indicates the difference between the transmission time of the RTT measurement signal and the reception time of the RTT response signal.

[0139] A UE-centric RTT positioning procedure is similar to the network-based procedure, except that the UE 1204 transmits uplink RTT measurement signal(s) (e.g., on resources allocated by the serving base station). The uplink RTT measurement signal(s) are measured by multiple network nodes 1202 in the neighborhood of the UE 1204. Each involved network node 1202 responds with a downlink RTT response signal and reports a network node Rx-Tx time difference measurement to the positioning entity. The network node Rx-Tx time difference measurement indicates the time difference between the arrival time of the RTT measurement signal at the network node 1202 and the transmission time of the RTT response signal. The UE 1204, if it is not the positioning entity, reports, for each network node 1202, a UE Rx-Tx time difference measurement that indicates the difference between the transmission time of the RTT measurement signal and the reception time of the RTT response signal.

[0140] In order to determine the location (x, y) of the UE 1204, the positioning entity needs to know the locations of the network nodes 1202, which may be represented in a reference coordinate system as (x_k, y_y), where k=1, 2, 3 in the example of FIG. 12. Where the UE 1204 is the positioning entity, a location server with knowledge of the network geometry (e.g., location server 230, LMF 270, SLP 272) may provide the locations of the involved network nodes 1202 to the UE 1204.

[0141] The positioning entity determines each distance 1210 (d_k, where k=1, 2, 3) between the UE 1204 and the respective network node 1202 based on the UE Rx-Tx and network node Rx-Tx time difference measurements and the speed of light, as described further below with reference to FIG. 13. Specifically, in the example of FIG. 12, the distance 1210-1 between the UE 1204 and the

network node **1202-1** is d_1 , the distance **1210-2** between the UE **1204** and the network node **1202-2** is d_2 , and the distance **1210-3** between the UE **1204** and the network node **1202-3** is d_3 . Once each distance **1210** is determined, the positioning entity can solve for the location (x, y) of the UE **1204** by using a variety of known geometric techniques, such as trilateration. From FIG. **12**, it can be seen that the location of the UE **1204** ideally lies at the common intersection of three semicircles, each semicircle being defined by radius d_k and center (x_k, y_k) , where $k=1, 2, 3$.

[0142] FIG. **13** is a diagram **1300** showing example timings of RTT measurement signals exchanged between a network node **1302** (labeled “Node”) and a UE **1304**, according to aspects of the disclosure. The UE **1304** may be any of the UEs described herein. The network node **1302** may be a base station (e.g., any of the base stations described herein), an RIS, a positioning beacon, another UE (e.g., connected over a sidelink), or the like.

[0143] In the example of FIG. **13**, the network node **1302** (labeled “BS”) sends an RTT measurement signal **1310** (e.g., PRS) to the UE **1304** at time T_1 . The RTT measurement signal **1310** has some propagation delay T_{Prop} as it travels from the network node **1302** to the UE **1304**. At time T_2 (the reception time of the RTT measurement signal **1310** at the UE **1304**), the UE **1304** measures the RTT measurement signal **1310**. After some UE processing time, the UE **1304** transmits an RTT response signal **1320** (e.g., SRS) at time T_3 . After the propagation delay T_{Prop} , the network node **1302** measures the RTT response signal **1320** from the UE **1304** at time T_4 (the reception time of the RTT response signal **1320** at the network node **1302**).

[0144] The UE **1304** reports the difference between time T_3 and time T_2 (i.e., the UE's **1304** Rx-Tx time difference measurement, shown as UE_Rx-Tx **1312**) to the positioning entity. Similarly, the network node **1302** reports the difference between time T_4 and time T_1 (i.e., the network node's **1302** Rx-Tx time difference measurement, shown as Node_Rx-Tx **1322**) to the positioning entity. Using these measurements and the known speed of light, the positioning entity can calculate the distance to the UE **1304** as $d = \frac{1}{2} * c * (\text{Node_Rx-Tx} - \text{UE_Rx-Tx}) = \frac{1}{2} * c * (T_4 - T_1) - \frac{1}{2} * c * (T_3 - T_2)$, where c is the speed of light.

[0145] Based on the known location of the network node **1302** and the distance between the UE **1304** and the network node **1302** (and at least two other network nodes **1302**), the positioning entity can calculate the location of the UE **1304**. As shown in FIG. **12**, the location of the UE **1304** lies at the common intersection of three semicircles, each semicircle being defined by a radius of the distance between the UE **1304** and a respective network node **1302**.

[0146] In an aspect, the positioning entity may calculate the UE's **1204/1304** location using a two-dimensional coordinate system; however, the aspects disclosed herein are not so limited, and may also be applicable to determining locations using a three-dimensional coordinate system, if the extra dimension is desired. Additionally, while FIG. **12** illustrates one UE **1204** and three network nodes **1202** and FIG. **13** illustrates one UE **1304** and one network node **1302**, as will be appreciated, there may be more UEs **1204/1304** and more network nodes **1202/1302**.

[0147] FIG. **14** is a diagram **1400** showing example timings of RTT measurement signals exchanged between a network node **1402** and a UE **1404**, according to aspects of the disclosure. The diagram **1400** is similar to the diagram **1300**, except that it includes processing delays that may occur at both the network node **1402** (labeled “Node”) and the UE **1404** when transmitting and receiving the RTT measurement and response signals. The network node **1402** may be a base station (e.g., any of the base stations), an RIS (e.g., RIS **410**), another UE (e.g., any of the UEs described herein), or other network node capable of performing an RTT positioning procedure. As a specific example, the network node **1402** and the UE **1404** may correspond to the network node **1302** and the UE **1304** in FIG. **13**.

[0148] Referring now to potential processing delays, at the network node **1402**, there is a transmission delay **1414** between the time T_1 that the network node's **1402** baseband (labeled “BB”) generates the RTT measurement signal **1410** (e.g., a PRS) and the time T_2 that the network node's **1402** antenna(s) (labeled “Ant”) transmit the RTT measurement signal **1410**. At the UE

1404, there is a reception delay **1416** between the time T_3 that the UE's **604** antenna(s) (labeled "Ant") receive the RTT measurement signal **1410** and the time T_4 that the UE's **1404** baseband (labeled "BB") processes the RTT measurement signal **1410**.

[0149] Similarly, for the RTT response signal **1420** (e.g., an SRS), there is a transmission delay **1426** between the time T_5 that the UE's **1404** baseband generates the RTT response signal **1420** and the time T_6 that the UE's **1404** antenna(s) transmit the RTT response signal **1420**. At the network node **1402**, there is a reception delay **1424** between the time T_7 that the network node's **1402** antenna(s) receive the RTT response signal **1420** and the time T_8 that the network node's **1402** baseband processes the RTT response signal **1420**.

[0150] The difference between times T_2 and T_1 (i.e., transmission delay **1414**) and times T_8 and T_7 (i.e., reception delay **1424**) is referred to as the network node's **1402** "group delay." The difference between times T_4 and T_3 (i.e., reception delay **1416**) and times T_6 and T_5 (i.e., transmission delay **1426**) is referred to as the UE's **1404** "group delay." The group delay includes a hardware group delay, a group delay attributable to software/firmware, or both. More specifically, although software and/or firmware may contribute to group delay, the group delay is primarily due to internal hardware delays between the baseband and the antenna(s) of the network node **1402** and the UE **1404**.

[0151] As shown in FIG. **14**, because of the reception delay **1416** and the transmission delay **1426**, the UE's **1404** Rx-Tx time difference measurement **1412** does not represent the difference between the actual reception time at time T_3 and the actual transmission time at time T_6 . Similarly, because of the transmission delay **1414** and the reception delay **1424**, the network node's **1402** Rx-Tx time difference measurement **1422** does not represent the difference between the actual transmission time at time T_2 and the actual reception time at time T_7 . Thus, as shown, group delays, such as reception delays **1416** and **1424** and transmission delays **1414** and **1426**, can contribute to timing errors and/or calibration errors that can impact RTT measurements, as well as other measurements, such as TDOA, RSTD, etc. This can in turn can impact positioning performance. For example, in some designs, a 10 ns error will introduce three meters of error in the final location estimate.

[0152] In some cases, the UE **1404** can calibrate its group delay and compensate for it so that the UE Rx-Tx time difference measurement **1412** reflects the actual reception and transmission times from its antenna(s). Alternatively, the UE **1404** can report its group delay to the positioning entity (if not the UE **1404**), which can then subtract the group delay from the UE Rx-Tx time difference measurement **1412** when determining the final distance between the network node **1402** and the UE **1404**. Similarly, the network node **1402** may be able to compensate for its group delay in the network node Rx-Tx time difference measurement **1422**, or simply report the group delay to the positioning entity.

[0153] FIG. **15** illustrates a time difference of arrival (TDOA)-based positioning procedure in an example wireless communications system **1500**, according to aspects of the disclosure. The TDOA-based positioning procedure may be an observed time difference of arrival (OTDOA) positioning procedure, as in LTE, or a downlink time difference of arrival (DL-TDOA) positioning procedure, as in 5G NR. In the example of FIG. **15**, a UE **1504** (e.g., any of the UEs described herein) is attempting to calculate an estimate of its location (referred to as "UE-based" positioning), or assist another entity (e.g., a base station or core network component, another UE, a location server, a third party application, etc.) to calculate an estimate of its location (referred to as "UE-assisted" positioning). The UE **1504** may communicate with (e.g., send information to and receive information from) one or more of a plurality of base stations **1502** (e.g., any combination of base stations described herein), labeled "BS1" **1502-1**, "BS2" **1502-2**, and "BS3" **1502-3**.

[0154] To support location estimates, the base stations **1502** may be configured to broadcast positioning reference signals (e.g., PRS, TRS, CRS, CSI-RS, etc.) to a UE **1504** in their coverage areas to enable the UE **1504** to measure characteristics of such reference signals. In a TDOA-based

positioning procedure, the UE **1504** measures the time difference, known as the reference signal time difference (RSTD) or TDOA, between specific downlink reference signals (e.g., PRS, TRS, CRS, CSI-RS, etc.) transmitted by different pairs of base stations **1502**, and either reports these RSTD measurements to a location server (e.g., location server **230**, LMF **270**, SLP **272**) or computes a location estimate itself from the RSTD measurements.

[0155] Generally, RSTDs are measured between a reference cell (e.g., a cell supported by base station **1502-1** in the example of FIG. **15**) and one or more neighbor cells (e.g., cells supported by base stations **1502-2** and **1502-3** in the example of FIG. **15**). The reference cell remains the same for all RSTDs measured by the UE **1504** for any single positioning use of TDOA and would typically correspond to the serving cell for the UE **1504** or another nearby cell with good signal strength at the UE **1504**. In an aspect, the neighbor cells would normally be cells supported by base stations different from the base station for the reference cell, and may have good or poor signal strength at the UE **1504**. The location computation can be based on the measured RSTDs and knowledge of the involved base stations' **1502** locations and relative transmission timing (e.g., regarding whether base stations **1502** are accurately synchronized or whether each base station **1502** transmits with some known time offset relative to other base stations **1502**).

[0156] To assist TDOA-based positioning operations, the location server (e.g., location server **230**, LMF **270**, SLP **272**) may provide assistance data to the UE **1504** for the reference cell and the neighbor cells relative to the reference cell. For example, the assistance data may include identifiers (e.g., PCI, VCI, CGI, etc.) for each cell of a set of cells that the UE **1504** is expected to measure (here, cells supported by the base stations **1502**). The assistance data may also provide the center channel frequency of each cell, various reference signal configuration parameters (e.g., the number of consecutive positioning slots, periodicity of positioning slots, muting sequence, frequency hopping sequence, reference signal identifier, reference signal bandwidth), and/or other cell related parameters applicable to TDOA-based positioning procedures. The assistance data may also indicate the serving cell for the UE **1504** as the reference cell.

[0157] In some cases, the assistance data may also include “expected RSTD” parameters, which provide the UE **1504** with information about the RSTD values the UE **1504** is expected to measure between the reference cell and each neighbor cell at its current location, together with an uncertainty of the expected RSTD parameter. The expected RSTD, together with the associated uncertainty, may define a search window for the UE **1504** within which the UE **1504** is expected to measure the RSTD value. In some cases, the value range of the expected RSTD may be ± 500 microseconds (μs). In some cases, when any of the resources used for the positioning measurement are in FR1, the value range for the uncertainty of the expected RSTD may be ± 32 μs . In other cases, when all of the resources used for the positioning measurement(s) are in FR2, the value range for the uncertainty of the expected RSTD may be ± 8 μs .

[0158] TDOA assistance information may also include positioning reference signal configuration information parameters, which allow the UE **1504** to determine when a positioning reference signal occasion will occur on signals received from various neighbor cells relative to positioning reference signal occasions for the reference cell, and to determine the reference signal sequence transmitted from the various cells in order to measure a reference signal time of arrival (ToA) or RSTD.

[0159] In an aspect, while the location server (e.g., location server **230**, LMF **270**, SLP **272**) may send the assistance data to the UE **1504**, alternatively, the assistance data can originate directly from the base stations **1502** themselves (e.g., in periodically broadcasted overhead messages, etc.). Alternatively, the UE **1504** can detect neighbor base stations itself without the use of assistance data.

[0160] The UE **1504** (e.g., based in part on the assistance data, if provided) can measure and (optionally) report the RSTDs between reference signals received from pairs of base stations **1502**. Using the RSTD measurements, the known absolute or relative transmission timing of each base

station **1502**, and the known location(s) of the reference and neighbor base stations **1502**, the network (e.g., location server **230**/LMF **270**/SLP **272**, a base station **1502**) or the UE **1504** can estimate the location of the UE **1504**. More particularly, the RSTD for a neighbor cell “k” relative to a reference cell “Ref” may be given as (ToA_k–ToA_{Ref}). In the example of FIG. **15**, the measured RSTDs between the reference cell of base station **1502-1** and the cells of neighbor base stations **1502-2** and **1502-3** may be represented as T2–T1 and T3–T1, where T1, T2, and T3 represent the ToA of a reference signal from the base station **1502-1**, **1502-2**, and **1502-3**, respectively. The UE **1504** (if it is not the positioning entity) may then send the RSTD measurements to the location server or other positioning entity. Using (i) the RSTD measurements, (ii) the known absolute or relative transmission timing of each base station **1502**, (iii) the known location(s) of the base stations **1502**, and/or (iv) directional reference signal characteristics, such as the direction of transmission, the UE's **1504** location may be determined (either by the UE **1504** or the location server).

[0161] In an aspect, the location estimate may specify the location of the UE **1504** in a two-dimensional (2D) coordinate system; however, the aspects disclosed herein are not so limited, and may also be applicable to determining location estimates using a three-dimensional (3D) coordinate system, if the extra dimension is desired. Additionally, while FIG. **15** illustrates one UE **1504** and three base stations **1502**, as will be appreciated, there may be more UEs **1504** and more base stations **1502**.

[0162] Still referring to FIG. **15**, when the UE **1504** obtains a location estimate using RSTDs, the necessary additional data (e.g., the base stations' **1502** locations and relative transmission timing) may be provided to the UE **1504** by the location server. In some implementations, a location estimate for the UE **1504** may be obtained (e.g., by the UE **1504** itself or by the location server) from RSTDs and from other measurements made by the UE **1504** (e.g., measurements of signal timing from global positioning system (GPS) or other global navigation satellite system (GNSS) satellites). In these implementations, known as hybrid positioning, the RSTD measurements may contribute towards obtaining the UE's **1504** location estimate but may not wholly determine the location estimate.

[0163] FIG. **16** is a diagram **1600** illustrating a base station (BS) **1602** (which may correspond to any of the base stations described herein) in communication with a UE **1604** (which may correspond to any of the UEs described herein). Referring to FIG. **16**, the base station **1602** may transmit a beamformed signal to the UE **1604** on one or more transmit beams **1602a**, **1602b**, **1602c**, **1602d**, **1602e**, **1602f**, **1602g**, **1602h**, each having a beam identifier that can be used by the UE **1604** to identify the respective beam. Where the base station **1602** is beamforming towards the UE **1604** with a single array of antennas (e.g., a single TRP/cell), the base station **1602** may perform a “beam sweep” by transmitting first beam **1602a**, then beam **1602b**, and so on until lastly transmitting beam **1602h**. Alternatively, the base station **1602** may transmit beams **1602a-1602h** in some pattern, such as beam **1602a**, then beam **1602h**, then beam **1602b**, then beam **1602g**, and so on. Where the base station **1602** is beamforming towards the UE **1604** using multiple arrays of antennas (e.g., multiple TRPs/cells), each antenna array may perform a beam sweep of a subset of the beams **1602a-1602h**. Alternatively, each of beams **1602a-1602h** may correspond to a single antenna or antenna array.

[0164] FIG. **16** further illustrates the paths **1612c**, **1612d**, **1612e**, **1612f**, and **1612g** followed by the beamformed signal transmitted on beams **1602c**, **1602d**, **1602e**, **1602f**, and **1602g**, respectively. Each path **1612c**, **1612d**, **1612e**, **1612f**, **1612g** may correspond to a single “multipath” or, due to the propagation characteristics of radio frequency (RF) signals through the environment, may be comprised of a plurality (a cluster) of “multipaths.” Note that although only the paths for beams **1602c-1602g** are shown, this is for simplicity, and the signal transmitted on each of beams **1602a-1602h** will follow some path. In the example shown, the paths **1612c**, **1612d**, **1612e**, and **1612f** are straight lines, while path **1612g** reflects off an obstacle **1620** (e.g., a building, vehicle, terrain

feature, etc.).

[0165] The UE **1604** may receive the beamformed signal from the base station **1602** on one or more receive beams **1604a**, **1604b**, **1604c**, **1604d**. Note that for simplicity, the beams illustrated in FIG. **16** represent either transmit beams or receive beams, depending on which of the base station **1602** and the UE **1604** is transmitting and which is receiving. Thus, the UE **1604** may also transmit a beamformed signal to the base station **1602** on one or more of the beams **1604a-1604d**, and the base station **1602** may receive the beamformed signal from the UE **1604** on one or more of the beams **1602a-1602h**.

[0166] In an aspect, the base station **1602** and the UE **1604** may perform beam training to align the transmit and receive beams of the base station **1602** and the UE **1604**. For example, depending on environmental conditions and other factors, the base station **1602** and the UE **1604** may determine that the best transmit and receive beams are **1602d** and **1604b**, respectively, or beams **1602e** and **1604c**, respectively. The direction of the best transmit beam for the base station **1602** may or may not be the same as the direction of the best receive beam, and likewise, the direction of the best receive beam for the UE **1604** may or may not be the same as the direction of the best transmit beam. Note, however, that aligning the transmit and receive beams is not necessary to perform a downlink angle-of-departure (DL-AoD) or uplink angle-of-arrival (UL-AoA) positioning procedure.

[0167] To perform a DL-AoD positioning procedure, the base station **1602** may transmit reference signals (e.g., PRS, CRS, TRS, CSI-RS, PSS, SSS, etc.) to the UE **1604** on one or more of beams **1602a-1602h**, with each beam having a different transmit angle. The different transmit angles of the beams will result in different received signal strengths (e.g., RSRP, RSRQ, SINR, etc.) at the UE **1604**. Specifically, the received signal strength will be lower for transmit beams **1602a-1602h** that are further from the line of sight (LOS) path **1610** between the base station **1602** and the UE **1604** than for transmit beams **1602a-1602h** that are closer to the LOS path **1610**.

[0168] In the example of FIG. **16**, if the base station **1602** transmits reference signals to the UE **1604** on beams **1602c**, **1602d**, **1602e**, **1602f**, and **1602g**, then transmit beam **1602e** is best aligned with the LOS path **1610**, while transmit beams **1602c**, **1602d**, **1602f**, and **1602g** are not. As such, beam **1602e** is likely to have a higher received signal strength at the UE **1604** than beams **1602c**, **1602d**, **1602f**, and **1602g**. Note that the reference signals transmitted on some beams (e.g., beams **1602c** and/or **1602f**) may not reach the UE **1604**, or energy reaching the UE **1604** from these beams may be so low that the energy may not be detectable or at least can be ignored.

[0169] The UE **1604** can report the received signal strength, and optionally, the associated measurement quality, of each measured transmit beam **1602c-1602g** to the base station **1602**, or alternatively, the identity of the transmit beam having the highest received signal strength (beam **1602e** in the example of FIG. **16**). Alternatively or additionally, if the UE **1604** is also engaged in a round-trip-time (RTT) or time-difference of arrival (TDOA) positioning session with at least one base station **1602** or a plurality of base stations **1602**, respectively, the UE **1604** can report reception-to-transmission (Rx-Tx) time difference or reference signal time difference (RSTD) measurements (and optionally the associated measurement qualities), respectively, to the serving base station **1602** or other positioning entity. In any case, the positioning entity (e.g., the base station **1602**, a location server, a third-party client, UE **1604**, etc.) can estimate the angle from the base station **1602** to the UE **1604** as the AoD of the transmit beam having the highest received signal strength at the UE **1604**, here, transmit beam **1602e**.

[0170] In one aspect of DL-AoD-based positioning, where there is only one involved base station **1602**, the base station **1602** and the UE **1604** can perform a round-trip-time (RTT) procedure to determine the distance between the base station **1602** and the UE **1604**. Thus, the positioning entity can determine both the direction to the UE **1604** (using DL-AoD positioning) and the distance to the UE **1604** (using RTT positioning) to estimate the location of the UE **1604**. Note that the AoD of the transmit beam having the highest received signal strength does not necessarily lie along the

LOS path **1610**, as shown in FIG. **16**. However, for DL-AoD-based positioning purposes, it is assumed to do so.

[0171] In another aspect of DL-AoD-based positioning, where there are multiple involved base stations **1602**, each involved base station **1602** can report, to the serving base station **1602**, the determined AoD from the respective base station **1602** to the UE **1604**, or the RSRP measurements. The serving base station **1602** may then report the AoDs or RSRP measurements from the other involved base station(s) **1602** to the positioning entity (e.g., UE **1604** for UE-based positioning or a location server for UE-assisted positioning). With this information, and knowledge of the base stations' **1602** geographic locations, the positioning entity can estimate a location of the UE **1604** as the intersection of the determined AoDs. There should be at least two involved base stations **1602** for a two-dimensional (2D) location solution, but as will be appreciated, the more base stations **1602** that are involved in the positioning procedure, the more accurate the estimated location of the UE **1604** will be.

[0172] To perform an UL-AoA positioning procedure, the UE **1604** transmits uplink reference signals (e.g., UL-PRS, SRS, DMRS, etc.) to the base station **1602** on one or more of uplink transmit beams **1604a-1604d**. The base station **1602** receives the uplink reference signals on one or more of uplink receive beams **1602a-1602h**. The base station **1602** determines the angle of the best receive beams **1602a-1602h** used to receive the one or more reference signals from the UE **1604** as the AoA from the UE **1604** to itself. Specifically, each of the receive beams **1602a-1602h** will result in a different received signal strength (e.g., RSRP, RSRQ, SINR, etc.) of the one or more reference signals at the base station **1602**. Further, the channel impulse response of the one or more reference signals will be smaller for receive beams **1602a-1602h** that are further from the actual LOS path between the base station **1602** and the UE **1604** than for receive beams **1602a-1602h** that are closer to the LOS path. Likewise, the received signal strength will be lower for receive beams **1602a-1602h** that are further from the LOS path than for receive beams **1602a-1602h** that are closer to the LOS path. As such, the base station **1602** identifies the receive beam **1602a-1602h** that results in the highest received signal strength and, optionally, the strongest channel impulse response, and estimates the angle from itself to the UE **1604** as the AoA of that receive beam **1602a-1602h**. Note that as with DL-AoD-based positioning, the AoA of the receive beam **1602a-1602h** resulting in the highest received signal strength (and strongest channel impulse response if measured) does not necessarily lie along the LOS path **1610**. However, for UL-AoA-based positioning purposes in FR2, it may be assumed to do so.

[0173] Note that while the UE **1604** is illustrated as being capable of beamforming, this is not necessary for DL-AoD and UL-AoA positioning procedures. Rather, the UE **1604** may receive and transmit on an omni-directional antenna.

[0174] Where the UE **1604** is estimating its location (i.e., the UE is the positioning entity), it needs to obtain the geographic location of the base station **1602**. The UE **1604** may obtain the location from, for example, the base station **1602** itself or a location server (e.g., location server **230**, LMF **270**, SLP **272**). With the knowledge of the distance to the base station **1602** (based on the RTT or timing advance), the angle between the base station **1602** and the UE **1604** (based on the UL-AoA of the best receive beam **1602a-1602h**), and the known geographic location of the base station **1602**, the UE **1604** can estimate its location.

[0175] Alternatively, where a positioning entity, such as the base station **1602** or a location server, is estimating the location of the UE **1604**, the base station **1602** reports the AoA of the receive beam **1602a-1602h** resulting in the highest received signal strength (and optionally strongest channel impulse response) of the reference signals received from the UE **1604**, or all received signal strengths and channel impulse responses for all receive beams **1602a-1602h** (which allows the positioning entity to determine the best receive beam **1602a-1602h**). The base station **1602** may additionally report the Rx-Tx time difference to the UE **1604**. The positioning entity can then estimate the location of the UE **1604** based on the UE's **1604** distance to the base station **1602**, the

AoA of the identified receive beam **1602a-1602h**, and the known geographic location of the base station **1602**.

[0176] High accuracy NR position estimation technologies (such as multi-RTT) require multiple positioning anchors (e.g., gNBs or anchor UEs). In particular, for timing measurement-based position estimation technologies, current NR technology requires multiple gNBs (or anchor UEs).

[0177] UEs may be classified as low-tier UEs (e.g., wearables, such as smart watches, glasses, rings, etc.) and premium UEs (e.g., smartphones, tablet computers, laptop computers, etc.). Low-tier UEs may alternatively be referred to as reduced-capability NR UEs, reduced-capability (RedCap) UEs. NR light UEs, light UEs, NR super light UEs, or super light UEs. Premium UEs may alternatively be referred to as full-capability UEs or simply UEs. Low-tier UEs generally have lower baseband processing capability, fewer antennas (e.g., one receiver antenna as baseline in FR1 or FR2, two receiver antennas optionally), lower operational bandwidth capabilities (e.g., 20 MHz for FR1 with no supplemental uplink or carrier aggregation, or 50 or 100 MHz for FR2), only half duplex frequency division duplex (HD-FDD) capability, smaller HARQ buffer, reduced physical downlink control channel (PDCCH) monitoring, restricted modulation (e.g., 64 QAM for downlink and 16 QAM for uplink), relaxed processing timeline requirements, and/or lower uplink transmission power compared to premium UEs. Different UE tiers can be differentiated by UE category and/or by UE capability. For example, certain types of UEs may be assigned a classification (e.g., by the original equipment manufacturer (OEM), the applicable wireless communications standards, or the like) of “low-tier” and other types of UEs may be assigned a classification of “premium.” Certain tiers of UEs may also report their type (e.g., “low-tier” or “premium”) to the network. Additionally, certain resources and/or channels may be dedicated to certain types of UEs.

[0178] As will be appreciated, the accuracy of low-tier UE positioning may be limited. For example, a low-tier UE may operate on a reduced bandwidth, such as 5 to 20 MHz for wearable devices and “relaxed” IoT devices (i.e., IoT devices with relaxed, or lower, capability parameters, such as lower throughput, relaxed delay requirements, lower energy consumption, etc.), which results in lower positioning accuracy. As another example, a low-tier UE's receive processing capability may be limited due to its lower cost RF/baseband. As such, the reliability of measurements and positioning computations would be reduced. In addition, such a low-tier UE may not be able to receive multiple PRS from multiple TRPs, further reducing positioning accuracy. As yet another example, the transmit power of a low-tier UE may be reduced, meaning there would be a lower quality of uplink measurements for low-tier UE positioning.

[0179] Premium UEs generally have a larger form factor and are costlier than low-tier UEs, and have more features and capabilities than low-tier UEs. For example, with respect to positioning, a premium UE may operate on the full PRS bandwidth, such as 100 MHz, and measure PRS from more TRPs than low-tier UEs, both of which result in higher positioning accuracy. As another example, a premium UE's receive processing capability may be higher (e.g., faster) due to its higher-capability RF/baseband. In addition, the transmit power of a premium UE may be higher than that of a low-tier UE. As such, the reliability of measurements and positioning computations would be increased.

[0180] Single-anchor (or single-gNB or single-anchor-UE) position estimation is a use-case for a coverage limited UE, such as low-tier UE position estimation. The low-tier UE could also achieve power saving through single-anchor position estimation. In some designs, hybrid position estimation such as RTT+AoA or RTT+AoD are promising single-anchor position estimation technologies. Hence angle estimation is required. However, high accuracy angle estimation accuracy may be challenging in multipath environment. For example, the performance of RTT+AoA method is quite sensitive to angle estimation error.

[0181] Aspects of the disclosure are directed to a reflection-based position estimation session. In some designs, the reflection-based position estimation session may be implemented between a UE

and a single anchor (e.g., gNB or anchor UE) via multipaths (or reflections) off of two (or more) reflectors. Such aspects may provide various technical advantages, such as facilitating higher accuracy position estimation in an environment (e.g., a multipath environment) where angle estimation accuracy is below a performance threshold. In particular, angle estimation need not be performed at all in accordance with some aspects of the reflection-based position estimation session.

[0182] FIG. 17 illustrates an exemplary process 1700 of communication according to aspects of the disclosure. The process 1700 of FIG. 1700 may be performed by a position estimation entity. In some designs, the position estimation entity may correspond to UE 302 (e.g., for UE-based position estimation) or a network component such as gNB or BS 304 (e.g., for UE-assisted position estimation with LMF integrated in RAN) or a core network component or location server such as network entity 306 (e.g., LMF).

[0183] Referring to FIG. 17, at 1710, the position estimation entity (e.g., receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, processor(s) 332 or 384 or 394, PRS component 342 or 388 or 398, etc.) determines a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node. For example, the wireless node may correspond to an anchor with a known location, such as gNB or UE anchor.

[0184] Referring to FIG. 17, at 1720, the position estimation entity (e.g., receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, processor(s) 332 or 384 or 394, PRS component 342 or 388 or 398, etc.) determines a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node. For example, the wireless node may correspond to an anchor with a known location, such as gNB or UE anchor.

[0185] Referring to FIG. 17, at 1730, the position estimation entity (e.g., receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, processor(s) 332 or 384 or 394, PRS component 342 or 388 or 398, etc.) determines a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path. The PRS configuration determined at 1730 may correspond to DL PRS configuration (e.g., wireless node is gNB and PRS is transmitted by gNB to UE), UL PRS configuration (e.g., wireless node is gNB and PRS is transmitted by UE to gNB), SL PRS configuration (e.g., wireless node is anchor UE and PRS is transmitted by anchor UE to UE and/or by UE to anchor UE), or a combination thereof (e.g. for two-way PRS procedure, such as RTT).

[0186] Referring to FIG. 17, at 1740, the position estimation entity (e.g., transmitter 314 or 324 or 354 or 364, network transceiver(s) 380 or 390, etc.) transmits an indication of the PRS configuration to the wireless node, the UE, or a combination thereof. For example, in some designs, the position estimation entity may correspond to the wireless node or the UE (e.g., PRS configuration need not be self-transmitted unless interpreted as an internal logical transfer of data) or a remote network component separate from the wireless node and UE.

[0187] Referring to FIG. 17, at 1750, the position estimation entity (e.g., receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, etc.) receives third measurement information associated with the position estimation session. Hence, the first and second locations of the first and second reflectors is determined, and then used to determine the PRS configuration for a position estimation session, with the third measurement information being associated with that position estimation session.

[0188] Referring to FIG. 17, at 1760, the position estimation entity (e.g., processor(s) 332 or 384 or 394, PRS component 342 or 388 or 398, etc.) derives a position estimate of the UE based in part

upon the first measurement information, the second measurement information, and the third measurement information.

[0189] FIG. **18** illustrates an exemplary process **1800** of communication according to aspects of the disclosure. The process **1800** of FIG. **1800** may be performed by a wireless node. For example, the wireless node may correspond to an anchor with a known location, such as gNB (e.g., BS **304**) or UE anchor (e.g., UE **302**).

[0190] Referring to FIG. **18**, at **1810**, the wireless node (e.g., receiver **312** or **322** or **352** or **362**, transmitter **314** or **324** or **354** or **364**, network transceiver(s) **380**, processor(s) **332** or **384**, PRS component **342** or **388**, etc.) performs a first sensing operation associated with a first reflector.

[0191] Referring to FIG. **18**, at **1820**, the wireless node (e.g., receiver **312** or **322** or **352** or **362**, transmitter **314** or **324** or **354** or **364**, network transceiver(s) **380**, processor(s) **332** or **384**, PRS component **342** or **388**, etc.) performs a second sensing operation associated with a second reflector.

[0192] Referring to FIG. **18**, at **1830**, the wireless node (e.g., transmitter **314** or **324** or **354** or **364**, network transceiver(s) **380**, etc.) reports, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation. For example, the position estimation entity may correspond to the position estimation entity that performs the process **1800** of FIG. **18**.

[0193] Referring to FIG. **18**, at **1840**, the wireless node (e.g., receiver **312** or **322** or **352** or **362**, transmitter **314** or **324** or **354** or **364**, network transceiver(s) **380**, etc.) receives, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path. The PRS configuration received at **1840** may correspond to DL PRS configuration (e.g., wireless node is gNB and PRS is transmitted by gNB to UE), UL PRS configuration (e.g., wireless node is gNB and PRS is transmitted by UE to gNB), SL PRS configuration (e.g., wireless node is anchor UE and PRS is transmitted by anchor UE to UE and/or by UE to anchor UE), or a combination thereof (e.g. for two-way PRS procedure, such as RTT).

[0194] Referring to FIG. **18**, at **1850**, the wireless node (e.g., receiver **312** or **322** or **352** or **362**, transmitter **314** or **324** or **354** or **364**, etc.) transmits or measures one or more PRSs to or from the UE in accordance with the PRS configuration. For example, the PRS(s) may be transmitted by the wireless node to the UE (e.g., DL PRS or SL PRS), or the PRS(s) may be received at the wireless node from the UE (e.g., UL PRS or SL PRS), or both (e.g., two-way PRS exchange for RTT measurement).

[0195] Referring to FIG. **18**, at **1860**, the wireless node (e.g., receiver **312** or **322** or **352** or **362**, transmitter **314** or **324** or **354** or **364**, network transceiver(s) **380**, processor(s) **332** or **384**, PRS component **342** or **388**, etc.) obtains third measurement information associated with the position estimation session. The third measurement information may be based on measurements performed by the wireless node itself (e.g., wireless node measures SL PRS or UL PRS from UE) or by the UE (e.g., UE measures SL PRS or DL PRS from wireless node, and then reports results back to wireless node) or both (e.g., two-way PRS exchange for RTT measurement).

[0196] FIG. **19** illustrates a position estimation environment **1900** in accordance with an example implementation of the processes **1700-1800** of FIGS. **17-18**, respectively. The position estimation environment **1900** includes UE **302** (i.e., target UE for position estimation) and BS **304** acting as a wireless node (or anchor). In other designs, as noted above, a UE anchor may be used in place of BS **304**. The position estimation environment **1900** further includes buildings **1910** and **1920**. In this example, buildings **1910** and **1920** may be characterized as reflectors. In FIG. **19**, a first PRS

1925 from BS **304** reaches UE **302** along an LOS path, a second PRS **1930** from BS **304** is reflected off building **1910**, and reaches UE **302** as a reflected PRS **1935**, and a second PRS **1940** from BS **304** is reflected off building **1920**, and reaches UE **302** as a reflected PRS **1945**. The first, second and third PRSs **1925**, **1930** and **1940** may correspond to the same PRS along different paths (e.g., multiple peaks for each path, as shown in FIG. **10**), or alternatively to different PRS (e.g., with a known offset between the transmission times of each PRS, with the measurement information being calibrated based on the known offset). Circumference **1950** has a radius with a distance corresponding to a sum of a distance of **1930** and **1935**, and circumference **1955** has a radius with a distance corresponding to a sum of a distance of **1940** and **1945**. While FIG. **19** depicts PRS in a direction from BS **304** to UE **302**, it will be appreciated that other aspects may involve PRS in a direction from UE **302** to BS **304** (or UE **302** to anchor UE, in case of sidelink implementation).

[0197] Referring to FIGS. **17-18**, in some designs, from receiver's (e.g., UE **302**) perspective, multiple peaks could be detected, which corresponding to multiple reflector as shown in FIG. **19**. To utilize the environment reflectors (buildings **1910** and **1920**) as the positioning anchors, the reflector locations should be estimated or referred. There could be various ways this can be achieved, such as RF sensing or a combination of RF sensing and a high precision map. In some designs, the wireless node (e.g., UE or gNB) may send the RF sensing results to the position estimation entity. The RF sensing measurement report may include the gNB/UE ID and time stamp. In some designs, the measurement may be the relative location of the reflector or some raw RF sensing measurement, such as RTT/angle estimation (for the mono-static radar sensing, or range sum/angle estimation with bi-static radar sensing).

[0198] Referring to FIGS. **17-18**, in some designs, to ensure the PRS experience the same multipath propagation as the radar sensing signal, various conditions may be implemented, including any of: [0199] Each PRS is associated with a specific radar RS (e.g., at least the PRS beam may be QCLed-C with radar RS), or [0200] The radar RS may be configured as similar as the PRS (e.g., the same or similar carrier frequency, the same or similar BW, closely scheduled in time, etc.), or, [0201] Cross validation of the estimated reflector locations (through RF sensing). For example, RTT could provide the range estimation between UE and gNB. The range sum (from gNB to reflector to UE) could be estimated through the relative delay and RTT. Here, range sum = (Relative delay)*c+RTT/2*c, whereby c is the speed of light. In some designs, the reflector should in the elliptic surface that is defined by the gNB-UE baseline (shown in FIGS. **19** as **1950** and **1955**). In some designs, the UE/gNB could indicate the cross-validation results to the location server, or UE/gNB only report the RF sensing results that passed the cross validation to the location server.

[0202] Referring to FIGS. **17-18**, in some designs, a PRS resource may be associated with one or multiple reflectors. For example, in the positioning assistance data, the LMF may indicate a specific PRS is preferred for the detection of a specific reflection. In some designs, multiple position estimation techniques may be used for multipath aided UE position estimation (e.g., TDOA-based position estimation and RTT-based position estimation).

[0203] For example, for TDOA-based position estimation, UE may report the RSTD derived from the relative delay between first arrival path and multipath. The position estimation entity may indicate which multipath could be used for the RSTD derivation. Each report may include the TRP ID/UE ID/reflector ID/PRS resource ID and time stamp. In an example, the RSTD may not be derived from a single channel estimate response (CER), since gNB may switch the Tx beams for the measurement of RSTD for different pair of UE/reflector. In some designs, as noted above, multiple PRS for different RSTD measurement should be scheduled closely in time to reduce the impact of timing drift.

[0204] In a further example, for RTT-based position estimation, UE and wireless node (e.g., gNB in case of BS anchor and another UE in case of UE anchor) may report multiple Rx-Tx time

difference to the position estimation entity. In some designs, a legacy Rx-Tx time difference report is based on the first arrival path detection in DL/UL (e.g., **1925** in FIG. **19**). In some designs, to reduce the overhead, the UE and/or wireless node may just report the additional Rx-Tx time difference offsets, which correspond to different multipath.

[0205] Referring to FIGS. **17-18**, in some designs, there may be a training phase to enable a long-term measurement-based reflector selection. For example, short-term channel variation may be due to mobile objects reflection/small scale fading/noise. Across a large number of measurements over a long-term training phase, only the common featured paths (reflector) are identified.

[0206] Referring to FIGS. **17-18**, in some designs, the optimum selection of reflectors could be location area-specific or UE-specific or some combination. For example, the channel itself varies across location. Different UEs' detection capability may also be different. For example, some UE may not be able to detect some weaker path, some UE may be able to detect the weaker path, and so on. Hence, the cross-validation mentioned above may be used.

[0207] Referring to FIGS. **17-18**, in some designs, the UE or the position estimation entity may request (e.g., on-demand) a re-selection of reflectors. For example, the optimum reflectors may change over time and UE locations.

[0208] Referring to FIGS. **17-18**, in some designs, after fusion of the RF sensing measurement and positioning measurement, the LMF will broadcast/unicast/group-cast the locations of reflector locations to the UEs. In some designs, the LMF may indicate whether a specific area is suitable for multipath aided position estimation. For example, in some environments, there may be no multipath (e.g., an unobstructed valley without candidate reflectors, etc.) or rich scattering, hence not suitable for the multi-path aided position estimation. In this case, LMF would not signal any additional assistance for multipath aided position estimation.

[0209] Referring to FIGS. **17-18**, in some designs, multipath aided UE position estimation could be applied for sidelink scenarios (e.g., wireless node is a UE rather than gNB). Since the position estimation anchor could be a mobile UE, the selection of reflectors could be more dynamic than the Uu interface-based position estimation. In some designs, whenever the anchor UE change its location (e.g., by more than some threshold), the re-selection of reflectors should be triggered. In some designs, even if the anchor UE's location does not change, the resection of reflectors may also need to be triggered (e.g., if its antenna orientation changes, or if performance is otherwise unsuitable, etc.).

[0210] Referring to FIGS. **17-18**, in some designs, the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0211] Referring to FIGS. **17-18**, in some designs, the position estimation entity may further receive information associated with a first PRS-based cross-validation procedure to verify the first location of the first reflector, a second PRS-based cross-validation procedure to verify the second location of second first reflector, or a combination thereof, and the first reflector and the second reflector are selected for participation in the position estimation session based on the information. For example, the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0212] Referring to FIGS. **17-18**, in some designs, a first set of PRS resources of the PRS configuration is associated with the first reflector, or a second set of PRS resources of the PRS

configuration is associated with the second reflector, or a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0213] Referring to FIGS. **17-18**, in some designs, the position estimation session comprises a time difference of arrival (TDOA) position estimation session. For example, the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS (e.g., the first PRS and the second PRS may be the same or different as noted above, e.g., same or different CERs) over the third path and the second PRS over the second path, or a combination thereof.

[0214] Referring to FIGS. **17-18**, in some designs, the position estimation session comprises a round trip time (RTT) position estimation session. For example, the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS (e.g., the first PRS and the second PRS may be the same or different as noted above, e.g., same or different CERs) at the wireless node or the UE over the second path, or a combination thereof.

[0215] Referring to FIGS. **17-18**, in some designs, the first reflector and the second reflector are selected from a group of reflectors based on one or more selection criteria. For example, the one or more selection criteria may include location area, UE capability, a request from the UE or the wireless node, or any combination thereof.

[0216] Referring to FIGS. **17-18**, in some designs, the position estimation entity further transmits a first indication of the first location of the first reflector, a second indication of the second location of the second reflector, or a combination thereof. In some designs, the position estimation entity further transmits an indication of whether a particular location area supports reflector-based position estimation.

[0217] Referring to FIGS. **17-18**, in some designs as noted above, the PRS configuration comprises configuration of one or more downlink PRSs, one or more uplink PRSs, one or more sidelink PRSs, or any combination thereof. In some designs, the wireless node corresponds to an anchor UE, and the PRS configuration includes at least the configuration of the one or more sidelink PRSs. In some designs, a new set of reflectors is selected for a new position estimation session of the UE in response to an anchor UE transition.

[0218] Referring to FIGS. **17-18**, in some designs, the wireless node may further transmits the third measurement information associated with the position estimation session to a position estimation entity, or the wireless node itself may derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information (e.g., for scenario where the wireless node itself is the position estimation entity).

[0219] Referring to FIGS. **17-18**, in some designs, the wireless node may further perform a first PRS-based cross-validation procedure to verify a first location of the first reflector, a second PRS-based cross-validation procedure to verify a second location of second first reflector, or a combination thereof, and the first measurement information and the second measurement information are reported based on the first PRS-based cross-validation procedure verifying the first location of the first reflector and the second PRS-based cross-validation procedure verifying the second location of the second reflector, or the first measurement information and the second measurement information are reported irrespective of whether the first PRS-based cross-validation procedure verifies the first location of the first reflector and the second PRS-based cross-validation procedure verifies the second location of the second reflector. For example, the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing

operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0220] Referring to FIGS. 17-18, in some designs, the wireless node may further receive a first indication of a first location of the first reflector, a second indication of a second location of the second reflector, or a combination thereof. In some designs, the wireless node may further receive an indication of whether a particular location area supports reflector-based position estimation.

[0221] In the detailed description above it can be seen that different features are grouped together in examples. This manner of disclosure should not be understood as an intention that the example clauses have more features than are explicitly mentioned in each clause. Rather, the various aspects of the disclosure may include fewer than all features of an individual example clause disclosed. Therefore, the following clauses should hereby be deemed to be incorporated in the description, wherein each clause by itself can stand as a separate example. Although each dependent clause can refer in the clauses to a specific combination with one of the other clauses, the aspect(s) of that dependent clause are not limited to the specific combination. It will be appreciated that other example clauses can also include a combination of the dependent clause aspect(s) with the subject matter of any other dependent clause or independent clause or a combination of any feature with other dependent and independent clauses. The various aspects disclosed herein expressly include these combinations, unless it is explicitly expressed or can be readily inferred that a specific combination is not intended (e.g., contradictory aspects, such as defining an element as both an insulator and a conductor). Furthermore, it is also intended that aspects of a clause can be included in any other independent clause, even if the clause is not directly dependent on the independent clause.

[0222] Implementation examples are described in the following numbered clauses:

[0223] Clause 1. A method of operating a position estimation entity, comprising: determining a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determining a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determining a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmitting an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receiving third measurement information associated with the position estimation session; and deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0224] Clause 2. The method of clause 1, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0225] Clause 3. The method of any of clauses 1 to 2, further comprising: receiving information associated with a first PRS-based cross-validation procedure to verify the first location of the first reflector, a second PRS-based cross-validation procedure to verify the second location of second first reflector, or a combination thereof, and wherein the first reflector and the second reflector are selected for participation in the position estimation session based on the information.

[0226] Clause 4. The method of clause 3, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0227] Clause 5. The method of any of clauses 1 to 4, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0228] Clause 6. The method of any of clauses 1 to 5, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0229] Clause 7. The method of clause 6, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0230] Clause 8. The method of any of clauses 1 to 7, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0231] Clause 9. The method of clause 8, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0232] Clause 10. The method of any of clauses 1 to 9, wherein the first reflector and the second reflector are selected from a group of reflectors based on one or more selection criteria.

[0233] Clause 11. The method of clause 10, wherein the one or more selection criteria comprises location area, UE capability, a request from the UE or the wireless node, or any combination thereof.

[0234] Clause 12. The method of any of clauses 1 to 11, further comprising: transmitting a first indication of the first location of the first reflector, a second indication of the second location of the second reflector, or a combination thereof.

[0235] Clause 13. The method of any of clauses 1 to 12, further comprising: transmitting an indication of whether a particular location area supports reflector-based position estimation.

[0236] Clause 14. The method of any of clauses 1 to 13, wherein the PRS configuration comprises configuration of one or more downlink PRSs, one or more uplink PRSs, one or more sidelink PRSs, or any combination thereof.

[0237] Clause 15. The method of clause 14, wherein the wireless node corresponds to an anchor UE, and wherein the PRS configuration comprises at least the configuration of the one or more sidelink PRSs.

[0238] Clause 16. The method of clause 15, wherein a new set of reflectors is selected for a new position estimation session of the UE in response to an anchor UE transition.

[0239] Clause 17. A method of operating a wireless node, comprising: performing a first sensing operation associated with a first reflector; performing a second sensing operation associated with a second reflector; reporting, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the

second sensing operation; receiving, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmitting or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtaining third measurement information associated with the position estimation session.

[0240] Clause 18. The method of clause 17, further comprising: transmitting the third measurement information associated with the position estimation session to a position estimation entity, or deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0241] Clause 19. The method of any of clauses 17 to 18, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0242] Clause 20. The method of any of clauses 17 to 19, further comprising: performing a first PRS-based cross-validation procedure to verify a first location of the first reflector, a second PRS-based cross-validation procedure to verify a second location of second first reflector, or a combination thereof, and wherein the first measurement information and the second measurement information are reported based on the first PRS-based cross-validation procedure verifying the first location of the first reflector and the second PRS-based cross-validation procedure verifying the second location of the second reflector, or wherein the first measurement information and the second measurement information are reported irrespective of whether the first PRS-based cross-validation procedure verifies the first location of the first reflector and the second PRS-based cross-validation procedure verifies the second location of the second reflector.

[0243] Clause 21. The method of clause 20, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0244] Clause 22. The method of any of clauses 17 to 21, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0245] Clause 23. The method of any of clauses 17 to 22, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0246] Clause 24. The method of clause 23, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0247] Clause 25. The method of any of clauses 17 to 24, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0248] Clause 26. The method of clause 25, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0249] Clause 27. The method of any of clauses 17 to 26, further comprising: receiving a first indication of a first location of the first reflector, a second indication of a second location of the second reflector, or a combination thereof.

[0250] Clause 28. The method of any of clauses 17 to 27, further comprising: receiving an indication of whether a particular location area supports reflector-based position estimation.

[0251] Clause 29. A position estimation entity, comprising: a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: determine a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determine a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determine a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit, via the at least one transceiver, an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receive, via the at least one transceiver, third measurement information associated with the position estimation session; and derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0252] Clause 30. The position estimation entity of clause 29, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0253] Clause 31. The position estimation entity of any of clauses 29 to 30, wherein the at least one processor is further configured to: receive, via the at least one transceiver, information associated with a first PRS-based cross-validation procedure to verify the first location of the first reflector, a second PRS-based cross-validation procedure to verify the second location of second first reflector, or a combination thereof, and wherein the first reflector and the second reflector are selected for participation in the position estimation session based on the information.

[0254] Clause 32. The position estimation entity of clause 31, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0255] Clause 33. The position estimation entity of any of clauses 29 to 32, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with

neither the first reflector nor the second reflector, or any combination thereof.

[0256] Clause 34. The position estimation entity of any of clauses 29 to 33, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0257] Clause 35. The position estimation entity of clause 34, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0258] Clause 36. The position estimation entity of any of clauses 29 to 35, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0259] Clause 37. The position estimation entity of clause 36, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0260] Clause 38. The position estimation entity of any of clauses 29 to 37, wherein the first reflector and the second reflector are selected from a group of reflectors based on one or more selection criteria.

[0261] Clause 39. The position estimation entity of clause 38, wherein the one or more selection criteria comprises location area, UE capability, a request from the UE or the wireless node, or any combination thereof.

[0262] Clause 40. The position estimation entity of any of clauses 29 to 39, wherein the at least one processor is further configured to: transmit, via the at least one transceiver, a first indication of the first location of the first reflector, a second indication of the second location of the second reflector, or a combination thereof.

[0263] Clause 41. The position estimation entity of any of clauses 29 to 40, wherein the at least one processor is further configured to: transmit, via the at least one transceiver, an indication of whether a particular location area supports reflector-based position estimation.

[0264] Clause 42. The position estimation entity of any of clauses 29 to 41, wherein the PRS configuration comprises configuration of one or more downlink PRSs, one or more uplink PRSs, one or more sidelink PRSs, or any combination thereof.

[0265] Clause 43. The position estimation entity of clause 42, wherein the wireless node corresponds to an anchor UE, and wherein the PRS configuration comprises at least the configuration of the one or more sidelink PRSs.

[0266] Clause 44. The position estimation entity of clause 43, wherein a new set of reflectors is selected for a new position estimation session of the UE in response to an anchor UE transition.

[0267] Clause 45. A wireless node, comprising: a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: perform a first sensing operation associated with a first reflector; perform a second sensing operation associated with a second reflector; report, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receive, via the at least one transceiver, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit, via the at least one transceiver, or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtain

third measurement information associated with the position estimation session.

[0268] Clause 46. The wireless node of clause 45, wherein the at least one processor is further configured to: transmit, via the at least one transceiver, the third measurement information associated with the position estimation session to a position estimation entity, or derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0269] Clause 47. The wireless node of any of clauses 45 to 46, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0270] Clause 48. The wireless node of any of clauses 45 to 47, wherein the at least one processor is further configured to: perform a first PRS-based cross-validation procedure to verify a first location of the first reflector, a second PRS-based cross-validation procedure to verify a second location of second first reflector, or a combination thereof, and wherein the first measurement information and the second measurement information are reported based on the first PRS-based cross-validation procedure verifying the first location of the first reflector and the second PRS-based cross-validation procedure verifying the second location of the second reflector, or wherein the first measurement information and the second measurement information are reported irrespective of whether the first PRS-based cross-validation procedure verifies the first location of the first reflector and the second PRS-based cross-validation procedure verifies the second location of the second reflector.

[0271] Clause 49. The wireless node of clause 48, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0272] Clause 50. The wireless node of any of clauses 45 to 49, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0273] Clause 51. The wireless node of any of clauses 45 to 50, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0274] Clause 52. The wireless node of clause 51, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0275] Clause 53. The wireless node of any of clauses 45 to 52, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0276] Clause 54. The wireless node of clause 53, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0277] Clause 55. The wireless node of any of clauses 45 to 54, wherein the at least one processor is further configured to: receive, via the at least one transceiver, a first indication of a first location of the first reflector, a second indication of a second location of the second reflector, or a combination thereof.

[0278] Clause 56. The wireless node of any of clauses 45 to 55, wherein the at least one processor is further configured to: receive, via the at least one transceiver, an indication of whether a particular location area supports reflector-based position estimation.

[0279] Clause 57. A position estimation entity, comprising: means for determining a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; means for determining a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; means for determining a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; means for transmitting an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; means for receiving third measurement information associated with the position estimation session; and means for deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0280] Clause 58. The position estimation entity of clause 57, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0281] Clause 59. The position estimation entity of any of clauses 57 to 58, further comprising: means for receiving information associated with a first PRS-based cross-validation procedure to verify the first location of the first reflector, a second PRS-based cross-validation procedure to verify the second location of second first reflector, or a combination thereof, and wherein the first reflector and the second reflector are selected for participation in the position estimation session based on the information.

[0282] Clause 60. The position estimation entity of clause 59, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0283] Clause 61. The position estimation entity of any of clauses 57 to 60, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0284] Clause 62. The position estimation entity of any of clauses 57 to 61, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0285] Clause 63. The position estimation entity of clause 62, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the

third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0286] Clause 64. The position estimation entity of any of clauses 57 to 63, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0287] Clause 65. The position estimation entity of clause 64, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0288] Clause 66. The position estimation entity of any of clauses 57 to 65, wherein the first reflector and the second reflector are selected from a group of reflectors based on one or more selection criteria.

[0289] Clause 67. The position estimation entity of clause 66, wherein the one or more selection criteria comprises location area, UE capability, a request from the UE or the wireless node, or any combination thereof.

[0290] Clause 68. The position estimation entity of any of clauses 57 to 67, further comprising: means for transmitting a first indication of the first location of the first reflector, a second indication of the second location of the second reflector, or a combination thereof.

[0291] Clause 69. The position estimation entity of any of clauses 57 to 68, further comprising: means for transmitting an indication of whether a particular location area supports reflector-based position estimation.

[0292] Clause 70. The position estimation entity of any of clauses 57 to 69, wherein the PRS configuration comprises configuration of one or more downlink PRSs, one or more uplink PRSs, one or more sidelink PRSs, or any combination thereof.

[0293] Clause 71. The position estimation entity of clause 70, wherein the wireless node corresponds to an anchor UE, and wherein the PRS configuration comprises at least the configuration of the one or more sidelink PRSs.

[0294] Clause 72. The position estimation entity of clause 71, wherein a new set of reflectors is selected for a new position estimation session of the UE in response to an anchor UE transition.

[0295] Clause 73. A wireless node, comprising: means for performing a first sensing operation associated with a first reflector; means for performing a second sensing operation associated with a second reflector; means for reporting, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; means for receiving, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; means for transmitting or measuring one or more PRSs with the UE in accordance with the PRS configuration; and means for obtaining third measurement information associated with the position estimation session.

[0296] Clause 74. The wireless node of clause 73, further comprising: means for transmitting the third measurement information associated with the position estimation session to a position estimation entity, or means for deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0297] Clause 75. The wireless node of any of clauses 73 to 74, wherein the first sensing operation,

the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0298] Clause 76. The wireless node of any of clauses 73 to 75, further comprising: means for performing a first PRS-based cross-validation procedure to verify a first location of the first reflector, a second PRS-based cross-validation procedure to verify a second location of second first reflector, or a combination thereof, and wherein the first measurement information and the second measurement information are reported based on the first PRS-based cross-validation procedure verifying the first location of the first reflector and the second PRS-based cross-validation procedure verifying the second location of the second reflector, or wherein the first measurement information and the second measurement information are reported irrespective of whether the first PRS-based cross-validation procedure verifies the first location of the first reflector and the second PRS-based cross-validation procedure verifies the second location of the second reflector.

[0299] Clause 77. The wireless node of clause 76, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0300] Clause 78. The wireless node of any of clauses 73 to 77, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0301] Clause 79. The wireless node of any of clauses 73 to 78, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0302] Clause 80. The wireless node of clause 79, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0303] Clause 81. The wireless node of any of clauses 73 to 80, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0304] Clause 82. The wireless node of clause 81, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0305] Clause 83. The wireless node of any of clauses 73 to 82. further comprising: means for receiving a first indication of a first location of the first reflector, a second indication of a second location of the second reflector, or a combination thereof.

[0306] Clause 84. The wireless node of any of clauses 73 to 83, further comprising: means for receiving an indication of whether a particular location area supports reflector-based position estimation.

[0307] Clause 85. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a position estimation entity, cause the position estimation entity

to: determine a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determine a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determine a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receive third measurement information associated with the position estimation session; and derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0308] Clause 86. The non-transitory computer-readable medium of clause 85, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0309] Clause 87. The non-transitory computer-readable medium of any of clauses 85 to 86, further comprising computer-executable instructions that, when executed by the position estimation entity, cause the position estimation entity to: receive information associated with a first PRS-based cross-validation procedure to verify the first location of the first reflector, a second PRS-based cross-validation procedure to verify the second location of second first reflector, or a combination thereof, and wherein the first reflector and the second reflector are selected for participation in the position estimation session based on the information.

[0310] Clause 88. The non-transitory computer-readable medium of clause 87, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0311] Clause 89. The non-transitory computer-readable medium of any of clauses 85 to 88, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0312] Clause 90. The non-transitory computer-readable medium of any of clauses 85 to 89, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0313] Clause 91. The non-transitory computer-readable medium of clause 90, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0314] Clause 92. The non-transitory computer-readable medium of any of clauses 85 to 91, wherein the position estimation session comprises a round trip time (RTT) position estimation

session.

[0315] Clause 93. The non-transitory computer-readable medium of clause 92, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0316] Clause 94. The non-transitory computer-readable medium of any of clauses 85 to 93, wherein the first reflector and the second reflector are selected from a group of reflectors based on one or more selection criteria.

[0317] Clause 95. The non-transitory computer-readable medium of clause 94, wherein the one or more selection criteria comprises location area, UE capability, a request from the UE or the wireless node, or any combination thereof.

[0318] Clause 96. The non-transitory computer-readable medium of any of clauses 85 to 95, further comprising computer-executable instructions that, when executed by the position estimation entity, cause the position estimation entity to: transmit a first indication of the first location of the first reflector, a second indication of the second location of the second reflector, or a combination thereof.

[0319] Clause 97. The non-transitory computer-readable medium of any of clauses 85 to 96, further comprising computer-executable instructions that, when executed by the position estimation entity, cause the position estimation entity to: transmit an indication of whether a particular location area supports reflector-based position estimation.

[0320] Clause 98. The non-transitory computer-readable medium of any of clauses 85 to 97, wherein the PRS configuration comprises configuration of one or more downlink PRSs, one or more uplink PRSs, one or more sidelink PRSs, or any combination thereof.

[0321] Clause 99. The non-transitory computer-readable medium of clause 98, wherein the wireless node corresponds to an anchor UE, and wherein the PRS configuration comprises at least the configuration of the one or more sidelink PRSs.

[0322] Clause 100. The non-transitory computer-readable medium of clause 99, wherein a new set of reflectors is selected for a new position estimation session of the UE in response to an anchor UE transition.

[0323] Clause 101. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a wireless node, cause the wireless node to: perform a first sensing operation associated with a first reflector; perform a second sensing operation associated with a second reflector; report, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receive, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtain third measurement information associated with the position estimation session.

[0324] Clause 102. The non-transitory computer-readable medium of clause 101, further comprising computer-executable instructions that, when executed by the wireless node, cause the wireless node to: transmit the third measurement information associated with the position estimation session to a position estimation entity, or derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

[0325] Clause 103. The non-transitory computer-readable medium of any of clauses 101 to 102, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.

[0326] Clause 104. The non-transitory computer-readable medium of any of clauses 101 to 103, further comprising computer-executable instructions that, when executed by the wireless node, cause the wireless node to: perform a first PRS-based cross-validation procedure to verify a first location of the first reflector, a second PRS-based cross-validation procedure to verify a second location of second first reflector, or a combination thereof, and wherein the first measurement information and the second measurement information are reported based on the first PRS-based cross-validation procedure verifying the first location of the first reflector and the second PRS-based cross-validation procedure verifying the second location of the second reflector, or wherein the first measurement information and the second measurement information are reported irrespective of whether the first PRS-based cross-validation procedure verifies the first location of the first reflector and the second PRS-based cross-validation procedure verifies the second location of the second reflector.

[0327] Clause 105. The non-transitory computer-readable medium of clause 104, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

[0328] Clause 106. The non-transitory computer-readable medium of any of clauses 101 to 105, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

[0329] Clause 107. The non-transitory computer-readable medium of any of clauses 101 to 106, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

[0330] Clause 108. The non-transitory computer-readable medium of clause 107, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

[0331] Clause 109. The non-transitory computer-readable medium of any of clauses 101 to 108, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

[0332] Clause 110. The non-transitory computer-readable medium of clause 109, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

[0333] Clause 111. The non-transitory computer-readable medium of any of clauses 101 to 110, further comprising computer-executable instructions that, when executed by the wireless node,

cause the wireless node to: receive a first indication of a first location of the first reflector, a second indication of a second location of the second reflector, or a combination thereof.

[0334] Clause 112. The non-transitory computer-readable medium of any of clauses 101 to 111, further comprising computer-executable instructions that, when executed by the wireless node, cause the wireless node to: receive an indication of whether a particular location area supports reflector-based position estimation.

[0335] Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0336] Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0337] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field-programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0338] The methods, sequences and/or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An example storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal (e.g., UE). In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

[0339] In one or more example aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other

medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0340] While the foregoing disclosure shows illustrative aspects of the disclosure, it should be noted that various changes and modifications could be made herein without departing from the scope of the disclosure as defined by the appended claims. The functions, steps and/or actions of the method claims in accordance with the aspects of the disclosure described herein need not be performed in any particular order. Furthermore, although elements of the disclosure may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated.

Claims

1. A method of operating a position estimation entity, comprising: determining a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determining a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determining a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmitting an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receiving third measurement information associated with the position estimation session; and deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.
2. The method of claim 1, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.
3. The method of claim 1, further comprising: receiving information associated with a first PRS-based cross-validation procedure to verify the first location of the first reflector, a second PRS-based cross-validation procedure to verify the second location of second first reflector, or a combination thereof, and wherein the first reflector and the second reflector are selected for participation in the position estimation session based on the information.
4. The method of claim 3, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the

second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.

5. The method of claim 1, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.

6. The method of claim 1, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.

7. The method of claim 6, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.

8. The method of claim 1, wherein the position estimation session comprises a round trip time (RTT) position estimation session.

9. The method of claim 8, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.

10. The method of claim 1, wherein the first reflector and the second reflector are selected from a group of reflectors based on one or more selection criteria.

11. The method of claim 10, wherein the one or more selection criteria comprises location area, UE capability, a request from the UE or the wireless node, or any combination thereof.

12. The method of claim 1, further comprising: transmitting a first indication of the first location of the first reflector, a second indication of the second location of the second reflector, or a combination thereof.

13. The method of claim 1, further comprising: transmitting an indication of whether a particular location area supports reflector-based position estimation.

14. The method of claim 1, wherein the PRS configuration comprises configuration of one or more downlink PRSs, one or more uplink PRSs, one or more sidelink PRSs, or any combination thereof.

15. The method of claim 14, wherein the wireless node corresponds to an anchor UE, and wherein the PRS configuration comprises at least the configuration of the one or more sidelink PRSs.

16. The method of claim 15, wherein a new set of reflectors is selected for a new position estimation session of the UE in response to an anchor UE transition.

17. A method of operating a wireless node, comprising: performing a first sensing operation associated with a first reflector; performing a second sensing operation associated with a second reflector; reporting, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receiving, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmitting or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtaining third measurement information associated with the position estimation session.

- 18.** The method of claim 17, further comprising: transmitting the third measurement information associated with the position estimation session to a position estimation entity, or deriving a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.
- 19.** The method of claim 17, wherein the first sensing operation, the second sensing operation, or both, comprise a radio frequency (RF) sensing operation associated with one or more radar reference signals or a Light Detection and Ranging (LIDAR) operation associated with one or more LIDAR signals.
- 20.** The method of claim 17, further comprising: performing a first PRS-based cross-validation procedure to verify a first location of the first reflector, a second PRS-based cross-validation procedure to verify a second location of second first reflector, or a combination thereof, and wherein the first measurement information and the second measurement information are reported based on the first PRS-based cross-validation procedure verifying the first location of the first reflector and the second PRS-based cross-validation procedure verifying the second location of the second reflector, or wherein the first measurement information and the second measurement information are reported irrespective of whether the first PRS-based cross-validation procedure verifies the first location of the first reflector and the second PRS-based cross-validation procedure verifies the second location of the second reflector.
- 21.** The method of claim 20, wherein the first PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a first PRS that is quasi-colocated (QCLed) with a first radar reference signal of the first sensing operation, and wherein the first PRS and the first radar reference signal are transmitted on a first bandwidth within a second window of time, or wherein the second PRS-based cross-validation procedure comprises transmission between the wireless node to the UE of a second PRS that is QCLed with a second radar reference signal of the second sensing operation, and wherein the second PRS and the second radar reference signal are transmitted on a second bandwidth within a second window of time, or a combination thereof.
- 22.** The method of claim 17, wherein a first set of PRS resources of the PRS configuration is associated with the first reflector, wherein a second set of PRS resources of the PRS configuration is associated with the second reflector, wherein a third set of PRS resources of the PRS configuration is associated with the first reflector and the second reflector, or wherein a fourth set of PRS resources of the PRS configuration is associated with neither the first reflector nor the second reflector, or any combination thereof.
- 23.** The method of claim 17, wherein the position estimation session comprises a time difference of arrival (TDOA) position estimation session.
- 24.** The method of claim 23, wherein the third measurement information comprises a first reference signal time difference (RSTD) between a first PRS over the third path and the first PRS over the first path, or wherein the third measurement information comprises a second reference signal time difference (RSTD) between a second PRS over the third path and the second PRS over the second path, or a combination thereof.
- 25.** The method of claim 17, wherein the position estimation session comprises a round trip time (RTT) position estimation session.
- 26.** The method of claim 25, wherein the third measurement information comprises a first receive-transmit (Rx-Tx) time difference relative to receipt of a first PRS at the wireless node or the UE over the first path, or wherein the third measurement information comprises a second Rx-Tx time difference relative to receipt of a second PRS at the wireless node or the UE over the second path, or a combination thereof.
- 27.** The method of claim 17, further comprising: receiving a first indication of a first location of the first reflector, a second indication of a second location of the second reflector, or a combination thereof.
- 28.** The method of claim 17, further comprising: receiving an indication of whether a particular

location area supports reflector-based position estimation.

29. A position estimation entity, comprising: a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: determine a first location of a first reflector based upon first measurement information associated with a first sensing operation by a wireless node; determine a second location of a second reflector based upon second measurement information associated with a second sensing operation by the wireless node; determine a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit, via the at least one transceiver, an indication of the PRS configuration to the wireless node, the UE, or a combination thereof; receive, via the at least one transceiver, third measurement information associated with the position estimation session; and derive a position estimate of the UE based in part upon the first measurement information, the second measurement information, and the third measurement information.

30. A wireless node, comprising: a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: perform a first sensing operation associated with a first reflector; perform a second sensing operation associated with a second reflector; report, to a position estimation entity, first measurement information associated with the first sensing operation and second measurement information associated with the second sensing operation; receive, via the at least one transceiver, in response to the report of the first measurement information and the second measurement information, a positioning reference signal (PRS) configuration associated with a position estimation session between the wireless node and a user equipment (UE), wherein the PRS configuration is associated with a first path between the wireless node and the UE associated with reflection off of the first reflector, a second path between the wireless node and the UE associated with reflection off of the second reflector, and a third path between the wireless node and the UE that is shorter than the first path and the second path; transmit, via the at least one transceiver, or measuring one or more PRSs with the UE in accordance with the PRS configuration; and obtain third measurement information associated with the position estimation session.
